

Everzol[®] ERC Solution

Everzol[®] ERC Dyes

永續染整解決方案

*Sustainability Solution for
Eco Resources Conservation*

高效能 更環保
Go Green Go Fast



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EVERZOL

中文版



I 概述

1. 應用範圍介紹

Everzol ERC 系列染料是建立於寬廣色域，含括極淺色系至極深色系，以及鮮豔色系到難染的綠色系。

此系列染料專為解決節能減廢而開發的新型反應性染料及相關解決方案，特點如下：

- (a) 淺中色系：高均染、高染色再現性，無光致變性組合以及高耐日光牢度組合
- (b) 中深色系：經濟型組合，高均染、高染色再現性
- (c) 鮮豔色系：高彩度，高牢度，符合品牌商需求
- (d) 極深色系：省水、節能、易洗、高牢度，可大幅減少水足跡及碳足跡
- (e) 難染綠色系：高均染，優異染色再現性，優異水洗／濕牢度

Everzol ERC 染料藉由以下優異染色特性，故可於染色達到以下優點：

- (a) 減少染料用量
- (b) 減少中性鹽用量
- (c) 高吸收、高固著
- (d) 水解少、易洗淨
- (e) 減少水洗次數
- (f) 降低皂洗溫度（從 98°C 降至 70°C）
- (g) 高均染，高染色再現性，減少重修風險
- (h) 總工序時間短，有效提升產能
- (i) 降低染色成本，減少水足跡及碳足跡

再者，Everzol ERC Solution 提供染整業保護生態及珍惜資源的具體可行方案，包含先進環保型染料及節能省水高效的水洗流程，達成：

- ✓ Go Green 達成對環境友善目標
- ✓ Go Fast 提升產品性能及滿足印染業需求

2. 染料儲存

Everzol ERC 染料建議儲存於乾燥和常溫的環境下，具有良好的儲存安定性。若是在高溫潮濕狀況長時間儲存，易造成部分染料水解及力度下降問題。

在現場使用時，一旦開封使用後，請將包裝袋密封及加蓋避免受潮。

3. 染色水質

建議使用軟水為宜。少量的鈣、鎂離子（一般在 100ppm 以下）可能於加鹼固色時，與鹼劑形成不溶性沉澱而導致低得色量、不均染和較低牢度。

微量的銅（2ppm 以上）、鐵（10ppm 以上）等金屬離子可能影響某些染料色光，建議加入合適的水質改善劑。而 EDTA（乙二胺四乙酸）類型的螯合劑會造成某些染料色相變化或者牢度不佳，故此不建議使用。

4. 染色用電解質的品質

染色用電解質建議使用芒硝或者精鹽，並要求鈣、鎂鹽含量低，避免在加鹼固著過程中造成染料析出色點及染色異常。

5. 染液的製備

- (a) Everzol ERC 染料具有極佳的溶解度。在現場之一般溶解方法為先用少量 40-50°C 溫水，將染料攪拌成均勻之糊狀，然後再加入 80-90°C 熱水並再攪拌均勻。溶解染料的水建議為處理後之軟水，pH 在 6-7 之間。
- (b) Everzol Turquoise Blue ERC 建議先用 60°C - 80°C 熱水溶解。

6. 紡織品的前處理

織物於染色前應進行前處理，以除去纖維上的雜質或漿料。工序的選擇則依被染物的純淨程度及所染顏色的鮮明度而定。

若一般使用雙氧水漂白後布面的活性氧必須適當去除。棉織物品經過絲光後，染料吸收效果更佳。前處理後織物具有良好吸濕性，且控制布面 pH 於中性。

6-1. 精練方式

除淺色及極鮮豔色系外，其餘深色或鈍色色系則只需施行精練工程，如下：

精練劑 2g/l

純鹼 2g/l 沸騰處理 95-98°C x 30 分鐘，然後以熱水及冷水洗淨，並酸中和以除去織物上雜質。

6-2. 漂白方式

欲達到鮮豔的色相及除去原棉上的色素及雜質，除精練工程外，可於同浴內合併漂白工程建議精練漂白方法：

材質：針織布（一浴精練漂）		材質：平織布（二浴精練漂）	
精練劑	0.5-2.0 g/l	(1) 退漿	
雙氧水 (35%)	5.0-10 %	精練劑	0.5-2.0 g/l
液鹼 (47°B'e)	2-4 g/l	退漿劑	2.0-3.0 g/l
雙氧水安定劑	0.5-1.0 g/l	98°C x 40-60'	
98°C x 40-60'		(2) 漂白	同針織布

7. 染色後處理工程

Everzol ERC 系列染料染色後，需經過充分的洗滌及皂煮工程，以去除未固著及水解染料。

Everzol ERC 系列染料藉高固著及水解染料少，故可將皂洗溫度從傳統染料習用之 98°C 降至 70°C，故可達到省水、節能、易洗、高牢度目標。建議使用具有螯合分散特點之陰離子高效型皂洗劑，不建議使用弱陽離子型皂洗劑以避免色花、浮色、不均染。並可依牢度需求提供以下三種染色後皂洗工程，並依客戶生產現況及設備進行調整。

步驟	MT (中溫洗淨)		MTS (中溫皂洗)		HT (高溫皂洗)	
1	50°C	酸中和	50°C	酸中和	50°C	酸中和
2	70°C	熱水洗	70°C	熱水洗	60°C	溫水洗
3	70°C	熱水洗	70°C	熱水洗	80°C	熱水洗
4	70°C	熱水洗	70°C	皂洗	98°C	皂洗
5	70°C	熱水洗	70°C	皂洗	80°C	熱水洗
6					60°C	溫水洗

附註：

- ① 若需更高牢度需求，建議後續加入簡易固色工程
- ② 出缸前 pH 維持 7.0 ± 0.5
- ③ 為提升翠綠、翠藍色系水洗牢度，推薦 Evertex CFIX-T 固色劑使用
- ④ 為加速洗淨流程，可於染色後先液流水洗 10 分鐘，再執行 MT/MTS/HT 洗程。
結束完整洗程後，加入一道冷水洗把染色物降溫後，出缸。

8. 鹽和鹼劑的推薦用法

推薦芒硝作為染色用電解質為宜；不推薦食鹽，因為內含雜質會造成染料凝集形成色點，同時可能腐蝕染色設備。通用的鹼劑多以純鹼最佳，為了確保染色穩定品質，染色後最終 pH 值應在 10.8-11.2 之間，可於染色最終段取樣及測量之。

以下為不同浴比下之鹽鹼推薦使用量，並請依現場與化驗室染色狀況採用一致染色條件，可以達到化驗室到現場優化之染色再現性。

浸染染色浴比低於 1:8（使用純鹼）								
染色濃度 %	<0.5%	0.5-1%	1-2%	2-3%	3-4%	4-5%	5-6%	≥6%
鹽 g/l	10	20	30	40	50	60	70	70
純鹼 g/l	12	18	22	22	22	22	22	22
染色時間(分)	30	45	60	60	60	60	60	60

浸染染色浴比高於 1:8（使用純鹼）								
染色濃度 %	<0.5%	0.5-1%	1-2%	2-3%	3-4%	4-5%	5-6%	≥6%
鹽 g/l	20	30	40	50	60	70	80	80
純鹼 g/l	10	15	20	20	20	20	20	20
染色時間(分)	30	45	60	60	60	60	60	60

浸染染色浴比低於 1:8（使用純鹼 + 液鹼）								
染色濃度 %	<0.5%	0.5-1%	1-2%	2-3%	3-4%	4-5%	5-6%	≥6%
鹽 g/l	10	20	30	40	50	60	70	70
純鹼 + 液鹼 g/l ml/l	5+0.6	5+0.6	5+1.1	5+1.1	5+1.6	5+1.6	5+2.2	5+2.2
染色時間(分)	30	45	60	60	60	60	60	60

浸染染色浴比高於 1:8（使用純鹼 + 液鹼）								
染色濃度 %	<0.5%	0.5-1%	1-2%	2-3%	3-4%	4-5%	5-6%	≥6%
鹽 g/l	20	30	40	50	60	70	80	80
純鹼 + 液鹼 g/l ml/l	5+0.5	5+0.5	5+1	5+1	5+1.5	5+1.5	5+2	5+2
染色時間(分)	30	45	60	60	60	60	60	60

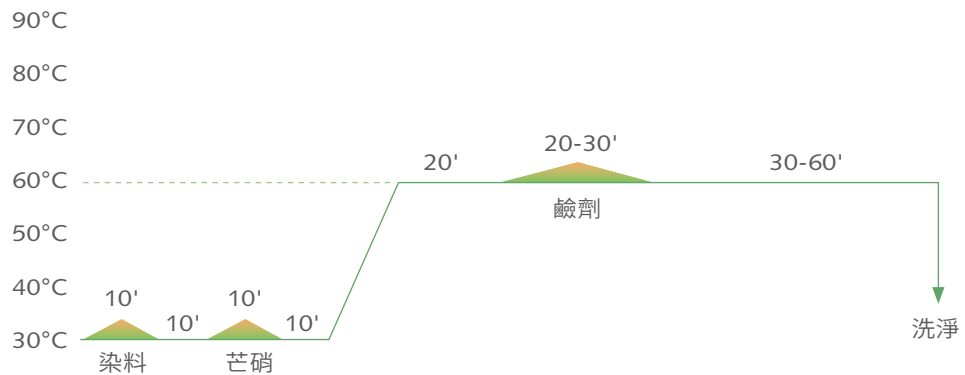
註：① 再生纖維素纖維，如 Viscose, Tencel, Modal 等及絲光棉，鹽用量可適量減少 20% 用量。

② Everzol Turquoise Blue ERC 建議使用芒硝染色。

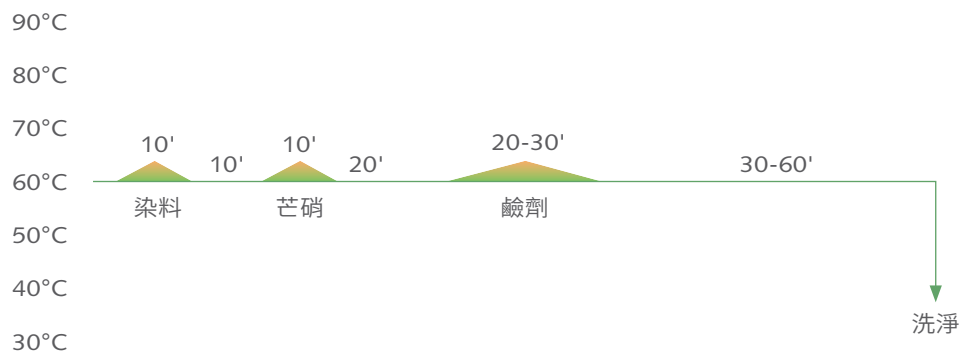
③ 此處使用之液鹼為 38° B'e

9. 染色方法介紹

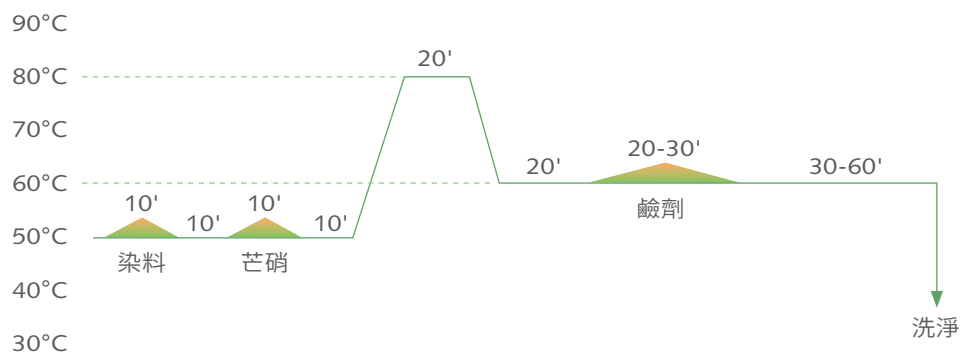
9-1. 升溫法



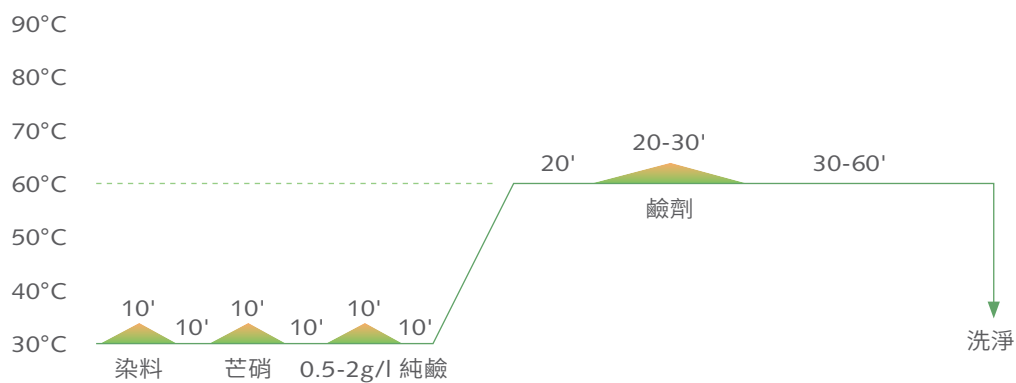
9-2. 恆溫法



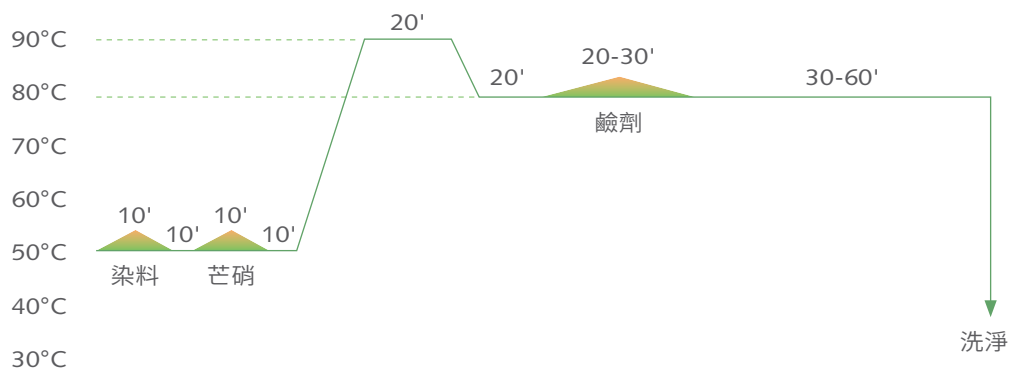
9-3. 高溫移染法 (用於 Viscose, Tencel, Modal 材質，或者厚重織物)



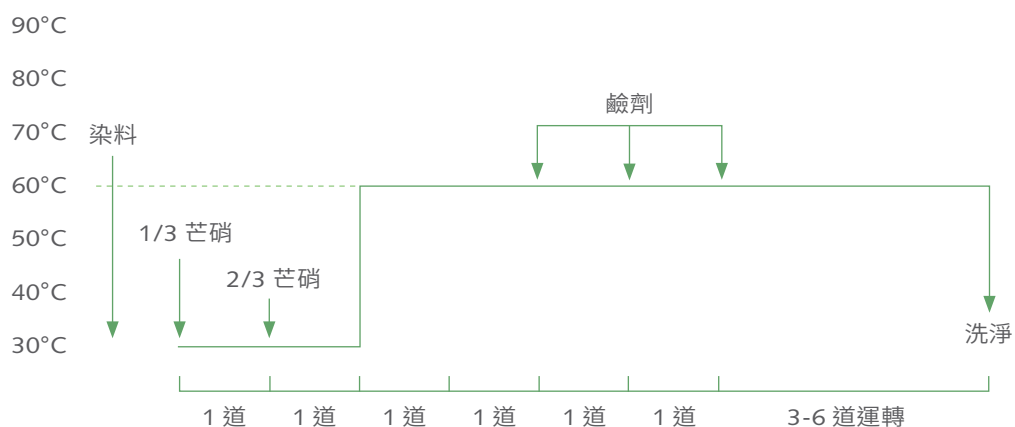
9-4. 預加鹼法



9-5. Everzol Turquoise Blue ERC 染色方法



9-6. Jigger 染色



10. Everzol ERC 染料修色

10-1. 不排液修色法

本方法推薦於修色量為原配方的 1% 以上

- (a) 染液冷卻至 40°C
- (b) 加入已稀釋溶解的染料
- (c) 在 40°C 持溫 10 分鐘
- (d) 升溫至 40°C
- (e) 60°C 持溫 30 分鐘
- (f) 對色

10-2. 排液修色法

此法用於淺色系或者敏感色系之修色

- (a) 排掉一半染液並加入冷水，可降低溫度及電解質濃度以避免色花
- (b) 加入已稀釋溶解染料
- (c) 在 40°C 持溫 10 分鐘
- (d) 升溫至 60°C
- (e) 60°C 持溫 30 分鐘
- (f) 對色

以上兩種修色方式，均無需添加芒硝及純鹼。

II 染色特性和堅牢度說明

基礎條件：

a 材質：未絲光針織布

b 助劑：芒硝、純鹼

c 染色溫度：(1) 一般色系 60°C

(2) Lemon ERC & Turquoise Blue ERC 80°C

d 染色濃度：一般色 3%，黑色 6%

e 染色浴比：1:10

1. 吸盡和固著曲線

評估：加入染料及中性鹽進行起始染色，升溫至 60°C，持溫 20 分鐘，加入鹼劑，固色 60 分鐘。

2. 反應性

加鹼反應時最高吸收率 pH 值的分類基準，分類如下：

pH 值	<11	I 類	高反應性
	11-11.5	II 類	中反應性
	>11.5	III 類	低反應性

3. 親合性

為鹽添加前染料吸收量的分類基準，分類如下：

吸收率	>75%	A 類	高親合性
	40-75%	B 類	中親合性
	<40%	C 類	低親合性

4. 移染性

先將已經一次吸盡染色的染色布與同材質的一塊白布放在浴比 1:10 並含有 60 g/l 芒硝的空白浴中，持溫 30 分鐘，加入 20 g/l 純鹼使兩塊布固色。

判定：對移染處理後的先染布樣和移染布樣（原白布），根據 AATCC 變褪色灰色標進行判定。

5. 洗淨性

先進行染色，染色濃度依照前述，並將染色後色樣控制壓吸率為 100%，然後依照以下水洗流程，並測定各工程的染料脫落量。

- | | |
|-----------------|---------------|
| (1) 60°C x 5 分鐘 | 溫水洗 |
| (2) 80°C x 5 分鐘 | 熱水洗 |
| (3) 98°C x 5 分鐘 | 皂洗（皂洗劑 3 g/l） |
| (4) 80°C x 5 分鐘 | 熱水洗 |
| (5) 60°C x 5 分鐘 | 溫水洗 |

$$\frac{\text{判定 (1) + (2) 染料脫落量}}{\text{全洗淨工程的脫落量}} \times 100\% = \text{洗淨性 (\%)}$$

洗淨性判定

96% 以上	5 級	71 - 75%	2-3 級
91 - 96%	4-5 級	66 - 70%	2 級
86 - 90%	4 級	60 - 65%	1-2 級
81 - 85%	3-4 級	60% 以下	1 級
76 - 80%	3 級		

6. 溶解度

(a) 不添加任何助劑

加入定量染料以 50°C 熱水溶解，攪拌 5 分鐘後過濾，根據過濾所得之染料殘留以判定染料溶解度。

(b) 加入 50 g/l 芒硝

加入定量染料及 50 g/l 芒硝，以 50°C 熱水溶解，攪拌 5 分鐘後冷卻到 25°C 過濾，根據過濾所得之染料殘留以判定染料溶解度。

7. 脫色性（浴比 1:10）

(1) 鹼處理

燒鹼（38°Bé）	20 ml/l
溫度 x 時間	80°C x 30 分鐘
溫水洗 ⇔ 冷水洗淨	

(2) 還原處理

燒鹼 (38°B'e) 20 ml/l
保險粉 5 g /l
溫度 x 時間 80°C x 30 分鐘
溫水洗 ⇨ 冷水洗淨

(3) 還原處理後再氧化處理

上述 (2) 條件處理後，再以下述氧化浴處理
次氯酸鈉 (有效氯 12%) 8 g/l
溫度 x 時間 30°C x 30 分鐘
溫水洗 ⇨ 冷水洗淨

(4) 氧化處理後再還原處理

以下列條件處理後，再以上述 (2) 的還原浴處理
次氯酸鈉 (有效氯 12%) 8 g/l
浴比 1:10
溫度 x 時間 30°C x 30 分鐘
溫水洗 ⇨ 冷水洗淨

判定：根據 AATCC 污染灰色標比較測試樣本與白胚布之差異

分類：(1) × 不適合 (含 2 級以下)

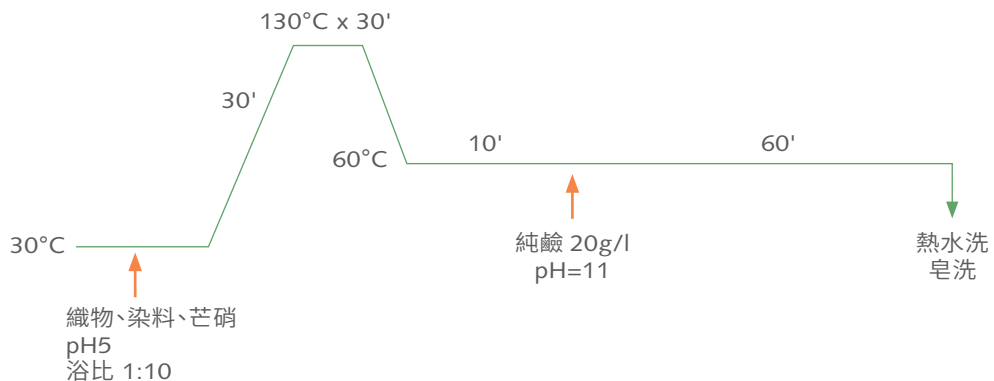
(2) △ 中等 (至 3 級)

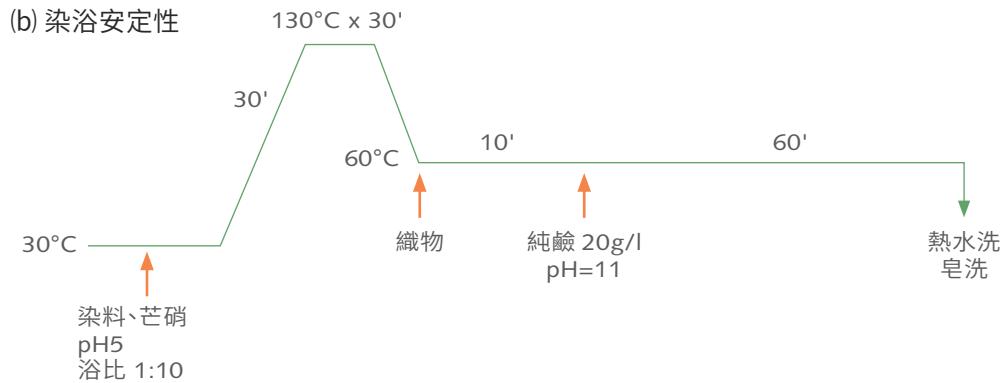
(3) ○ 適合 (含 3-4 級以上)

8. 高溫染色安定性

(a) 染色布之安定性

依前述測試染色濃度染色後的色布為基準，與以下列方法染色後的色布進行濃度及色光的變化，來評價於聚酯／棉混紡品染色性。





判定：以原樣為標準，再與測試樣本以灰色標評估色變

9. 色恆性

以標準光源 (D65) 作為基準，將在鎢絲燈 (Incandescent) 和超效能日光燈 (TL-84) 下的色光變化傾向說明：

Y：黃色調 R：紅色調 B：藍色調 G：綠色調

10. 染色浴中金屬離子對色光影響

(a) Fe^{2+} ：使用 FeCl_2 ，20ppm

(b) Cu^{2+} ：使用 CuSO_4 ，20ppm

判定：以原樣為標準，再與測試樣本以灰色標評估色變

11. 固色劑種類變色性

(a) 聚乙烯聚胺系：2% o.w.f 處理 60°C x 20 分鐘

(b) 四級胺鹽系：2% o.w.f 處理 60°C x 20 分鐘

判定：以未固色樣為標準，再與固色處理後樣本以灰色標評估色變

12. 樹酯加工性

(1) 樹酯加工處方：

A. 三聚氰胺樹脂

Resin M-3	40 g
Catalyst MRK	8 g
水	X g

全量	1000 g
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B. 乙二醛樹脂

Resin GF-4000	40 g
Catalyst G	8 g
水	X g

全量	1000 g
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(2) 處理：壓吸（壓吸率 70%）➡ 乾燥（100°C x 3 分鐘）➡ 烘乾（160°C x 3 分鐘）

判定：以原樣為標準，再與測試樣本以灰色標評估色變

13. 拔染性

染色後色布用下述二種處方進行拔染性試驗。

(1) 拔染糊處方：

	中性	鹼性
拔染劑（Rongalit C）	200	200
純鹼	—	50
吸濕劑	X	X
糊劑	Y	Y
水	Z	Z
全量	1000 g	1000 g

(2) 工程：印花 ➡ 100°C 烘乾 ➡ 102°C x 5~8 分鐘蒸處 ➡ 水洗及皂洗

14. 不同染色溫度依存性（加鹼後固色時間 60 分鐘）

染色溫度 (°C)	40	50	60	70	80	90	98
一般染料	✓	✓	✓	✓	✓		
Lemon/Turquoise 染料			✓	✓	✓	✓	✓

15. 鹽類用量依存性（加鹼後固色時間 60 分鐘）

染色濃度	請參考 Everzol ERC 系列染料技術資料					
鹽用量 (g /l)	30	40	50	60	70	80

16. 染色浴比依存性（浴比 1:10 為標準）

染色濃度	請參考 Everzol ERC 系列染料技術資料										
浴 比	5	8	10	12	15	20	30	40	60	80	120

17. 鹼劑種類與用量依存性（以 20 g/l 純鹼為標準）

染色濃度	請參考 Everzol ERC 系列染料技術資料					
鹼劑種類及用量	(a)	(b)	(c)	(d)	(e)	(f)
純鹼 (g/l)	5	10	15	20	5	5
液鹼 (38°B° e) ml/l	-	-	-	-	1	2

18. 染色堅牢度的試驗方法

- | | |
|---------------|---|
| 1. 耐水洗 | ISO 105 C10 方法 B
ISO 105 C10 方法 C
ISO 105 C10 方法 D
ISO 105 C06-C15
ISO 105 C06-C25
Marks & Spencer C4A |
| 2. 耐水 | ISO 105 E01 |
| 3. 耐熱水 | ISO 105 E08 |
| 4. 耐海水 | ISO 105 E02 |
| 5. 耐後絲光 | ISO 105 X04 |
| 6. 耐酸水解 | 1% 醋酸溶液浸泡 30 分鐘，進行 38°C x 6 小時處理 |
| 7. 耐汗（酸、鹼） | ISO 105 E04 |
| 8. 耐氯水（20ppm） | ISO 105 E03 |
| 9. 雙氧水漂白 | ISO 105 N02 |
| 10. 耐磨擦（乾、濕） | ISO 105 X12 |
| 11. 擬溼日光 | Marks & Spencer C9A |
| 12. 耐多次水洗 | ISO 105 C09 |
| 13. 耐唾液 | DIN 53160 |
| 14. 耐熱壓 | ISO 105 X11 |
| 15. 日光堅牢度 | ISO 105 B02 & AATCC 16-3 |
| 16. 汗日光堅牢度 | ISO 105 B07 & AATCC 125 |
| 17. 氣體變褪色 | ISO 105-G01 |

臺灣永光化學工業股份有限公司
Everlight Chemical Industrial Corporation

EVERZOL

English Version



I Overview

1. Application for exhaust dyeing

Everzol ERC series dyes are based on a wide color gamut, including very light colors to very dark colors, and bright colors to critical green colors. Such innovative dyes and related eco wash-off are specially developed to solve energy conservation and waste reduction. The characteristics are as follows:

- (a) pale-medium shades: unique levelling, excellent Right-First-Time, non-photochromism and high lightfastness.
- (b) medium-dark shades: cost-effective, unique levelling, excellent Right-First-Time.
- (c) brilliant shades: high chromaticity, high fastness, meet apparel brands demands.
- (d) very dark shades: save water and energy, easy wash-off, unique fastness, reduce water and CO₂ footprint.
- (e) critical green shades: unique levelling, excellent Right-First-Time, unique washing/wet fastness.

Thanks for the remarkable dyeing property of Everzol ERC dyes, the following benefits had been achieved:

- (a) reduce dyes consumption
- (b) reduce salt usage
- (c) high exhaustion and high fixation
- (d) less hydrolyzation and easy wash-off
- (e) reduce numbers of wash-off
- (f) reduce soaping temperature from 98°C to 70°C
- (g) unique levelling and Right-First-Time, less reprocess risk
- (h) compact total process time, increased productivity
- (i) reduce total cost, reduce water and CO₂ footprint

Additionally, Everzol ERC Solution, Eco Resources Conservation concept, provides the practical environmental friendly concrete action through the matrix of innovative dyes and revolutionary wash-off process to achieve:

- ✓ Go Green to achieve eco-friendly goal.
- ✓ Go Fast to enhance highest satisfactions for dyeing mills.

2. Dyes Storage

Powdered Everzol ERC dye are stable under the dry and room temperature storage condition to avoid dyes hydrolyzation and less color-yield. Sealed package after opening of operation in production site.

3. Water Quality

Softening and demineralized water should be used to avoid precipitation in alkaline fixation caused by Calcium and Magnesium ions (less than 100 ppm), along with less color yield, unleveling, and inferior fastness.

The addition of water softening agent will minimize the shade variation caused by very small amount of Copper (above 2ppm) and Iron (above 10ppm) ions. EDTA (Ethylene Diamine Tetra Acetic) is not recommended owing to cause shade change or impaired fastness on certain dyes.

4. Electrolyte Quality

Demanding less Calcium and Magnesium ions contained in electrolyte to avoid dyes precipitation and defectives in alkali addition. Glauber's salt and refined common salt will be recommended.

5. Preparation of Dyes Solution

- (a) Everzol ERC dyes possess good solubility. To completely dissolve in water pasted dyes with a small amount of warm water (40-50°C) with thorough stirring. Then hot water (80-90°C) and mix thoroughly. pH ranged at 6-7 of Softening water is essential.
- (b) Dissolving Everzol Turquoise Blue ERC with 60-80°C hot water.

6. Pretreatment

Proper pretreatment of impurity and sizings before dyeing will affect the dyeing quality. The original condition of gray materials and final shade required will rely on the proper selection of pretreatment.

The proper removal of oxygen residual after pretreatment should be carried out. The mercerization will enhance the dyes absorption. It is advisable to accentuate the capillarity and neutral pH of treated materials before dyeing.

6-1. Scouring process

The dark or dull shades, except pale and brilliant shades, may proceed the scouring process as below:

Scouring agent 2 g/l

Soda ash 2 g/l boiling at 95-98°C for 30 mins, then following hot and cold rinsing after neutralization to remove impurity.

6-2. Bleaching process

The same bath of scouring and bleaching will achieve the brilliant shades dyeing and remove the inherent coloured impurities from cotton cultivation.

Recommended scouring and bleaching process:

Knits Cotton (one bath)		Woven Cotton (two baths)	
Scouring agent	0.5-2.0 g/l	(1) Desizing	
H ₂ O ₂ (35%)	5.0-10 %	Scouring agent	0.5-2.0 g/l
Caustic Soda (47°B'e)	2-4 g/l	Desizing agent	2.0-3.0 g/l
Peroxide Stabilizer	0.5-1.0 g/l	(2) Bleaching	same as knits material
98°C x 40-60'		98°C x 40-60'	

7. Wash-Off Process after Dyeing

The additional wash-off and soaping after dyeing required for removing hydrolyzed and unfixed dyes .

The benefits of less hydrolyzed dyes and high fixation of Everzol ERC dyes will possibly reduce the soaping temperature from 98°C to 70°C to achieve the total savings in terms of water, energy, process time, even high fastness performance. Those anionic soaping agents with high dispersing properties will be highly recommended, instead of cationic typed soaping agent may cause unleveling and patching.

The below 3 eco wash-off processes for Everzol ERC dyes are advisable for existing exhaust dyeing facilities as below:

Step	MT (moderate temperature rinsing)		MTS (moderate temperature soaping)		HT (high temperature soaping)	
1	50°C	neutralizing	50°C	neutralizing	50°C	neutralizing
2	70°C	hot rinsing	70°C	hot rinsing	60°C	warm rinsing
3	70°C	hot rinsing	70°C	hot rinsing	80°C	hot rinsing
4	70°C	hot rinsing	70°C	soaping	98°C	soaping
5	70°C	hot rinsing	70°C	soaping	80°C	hot rinsing
6	--	--	--	--	60°C	warm rinsing

Remark :

- ① *the additional fixing after treatment will be required for higher washing fastness.*
- ② *pH controlling at 7.0 ± 0.5 before unloading.*
- ③ *Evertex CFIX-T will promote the unique washing fastness for green and turquoise blue shades.*
- ④ *Over-flow rinsing by 10 minutes after dyeing will be preferred before starting MT/MTS/HT wash-off. Additional cold rinsing is needed before unloading after complete wash-off to cool down the substrate temperature.*

8. Salt and Alkali Required of Everzol ERC dyes:

Glauber's salt anhydrous will be highly recommended. Common salt with high concentrated Calcium and Magnesium may cause dyes precipitation, even lead to vessel erosion.

Soda ash are recommended, even mixed with caustic soda, care must be taken with

industrial grades to minimize the risk of lower pH. The final dyeing pH after alkali addition by sampling and measuring should be ranged from 10.8-11.2.

Furthermore, salt and alkali addition may vary from different dyeing facilities. The same dyeing condition of liquor ratio and salt/alkali dosage will improve the Right-First-Time from lab to bulk and bulk to bulk consistency.

Liquor ratio below 1:8 (use soda ash only)								
Depth%	<0.5%	0.5-1%	1-2%	2-3%	3-4%	4-5%	5-6%	≥6%
Electrolyte g/l	10	20	30	40	50	60	70	70
Soda ash g/l	12	18	22	22	22	22	22	22
Fixation time (min)	30	45	60	60	60	60	60	60

Liquor ratio above 1:8 (use soda ash only)								
Depth%	<0.5%	0.5-1%	1-2%	2-3%	3-4%	4-5%	5-6%	≥6%
Electrolyte g/l	20	30	40	50	60	70	80	80
Soda ash g/l	10	15	20	20	20	20	20	20
Fixation time (min)	30	45	60	60	60	60	60	60

Liquor ratio below 1:8 (use mixed soda ash and caustic soda)								
Depth%	<0.5%	0.5-1%	1-2%	2-3%	3-4%	4-5%	5-6%	≥6%
Electrolyte g/l	10	20	30	40	50	60	70	70
Soda ash+ caustic soda g/l ml/l	5+0.6	5+0.6	5+1.1	5+1.1	5+1.6	5+1.6	5+2.2	5+2.2
Fixation time (min)	30	45	60	60	60	60	60	60

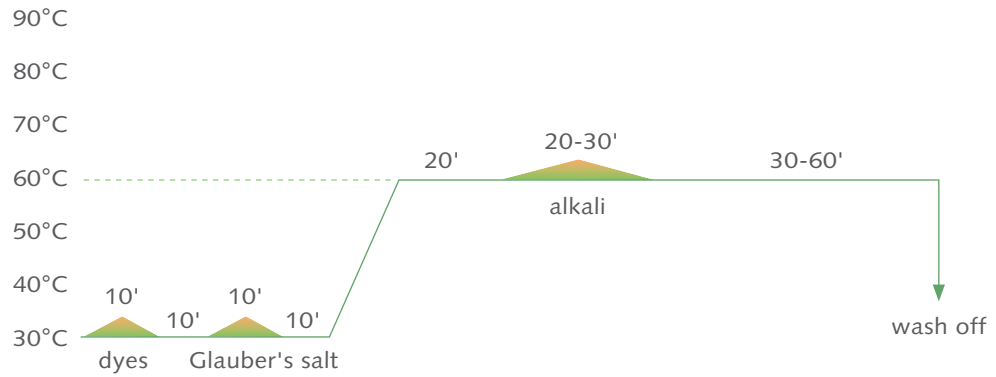
Liquor ratio above 1:8 (use mixed soda ash and caustic soda)								
Depth%	<0.5%	0.5-1%	1-2%	2-3%	3-4%	4-5%	5-6%	≥6%
Electrolyte g/l	20	30	40	50	60	70	80	80
Soda ash+ caustic soda g/l ml/l	5+0.5	5+0.5	5+1	5+1	5+1.5	5+1.5	5+2	5+2
Fixation time (min)	30	45	60	60	60	60	60	60

Remark:

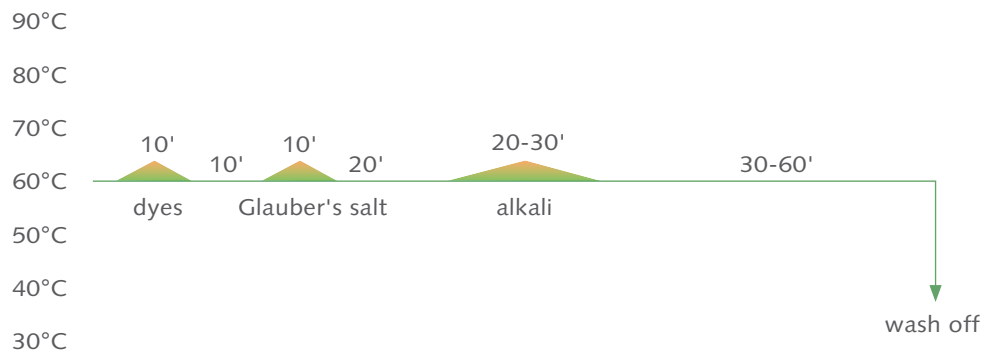
- ① 20% less salt used will be applied in regenerated cellulosic materials, like as viscose, Tencel, Modal, and mercerized cotton.
- ② Glauber's salt will be highly recommended for dyeing Everzol Turquoise Blue ERC.
- ③ Caustic Soda 38° B'e.

9. Dyeing Method

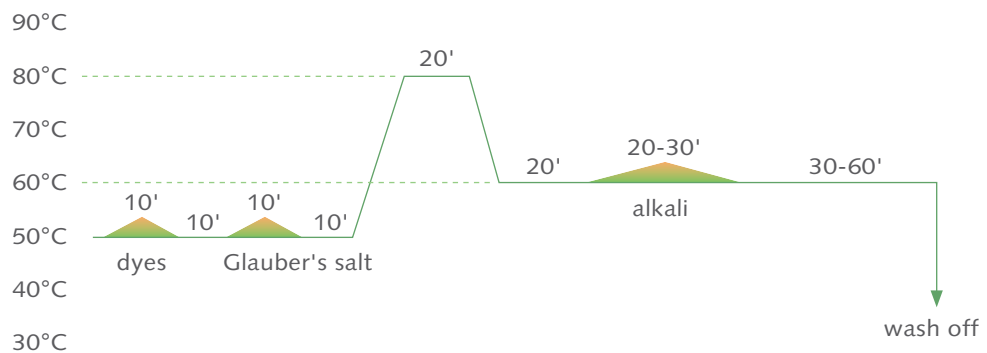
9-1. Temperature rise method



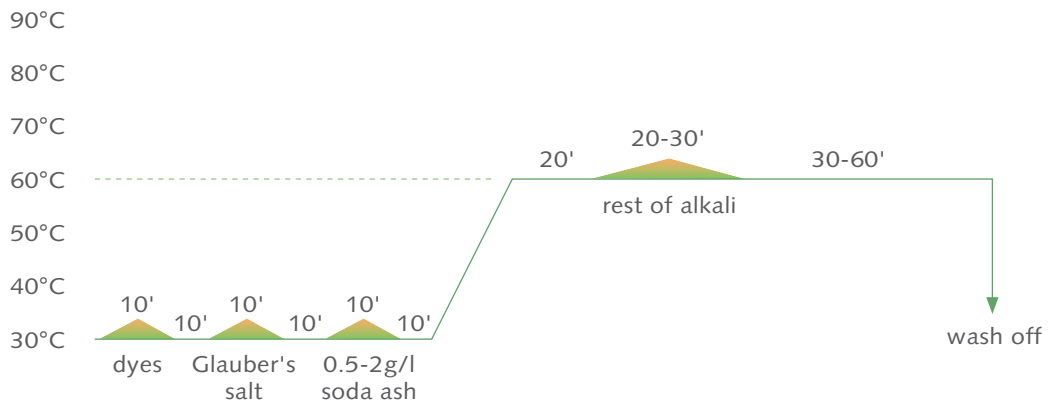
9-2. Isothermal method



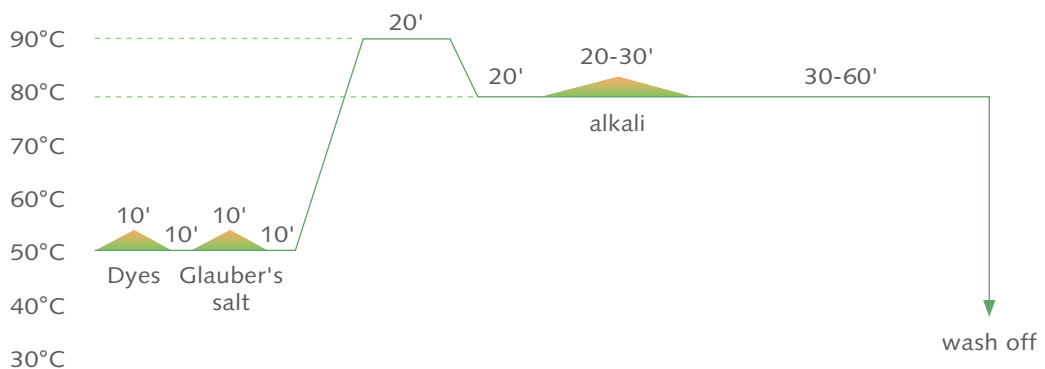
9-3. Migration (for Viscose, Tencel, Modal, or heavy fabrics)



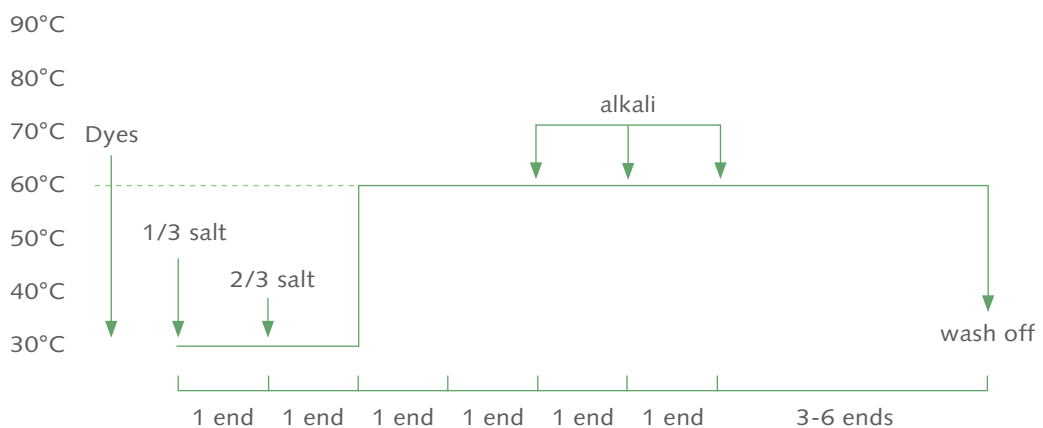
9-4. Semi-All-In method



9-5. Everzol Turquoise Blue ERC dyeing



9-6. Jigger dyeing



10. Shade correction for Everzol ERC dyes

10-1. Without draining bath method for slight shading

Recommended when the color correction amount is 1% or more of the original recipe

- (a) Cool dye bath to 40°C
- (b) Add diluted dyes
- (c) Hold 10 min at 40°C
- (d) Heat to 60°C
- (e) Hold 30 min at 60°C
- (f) Sampling and check shades

10-2. Draining partial bath method

Recommended for pale and critical shades correction

- (a) Drain half bath and add cold water to cool temperature and reduce the concentration of electrolyte to avoid unlevelling
- (b) Add diluted dyes
- (c) Hold 10 min at 40°C
- (d) Heat to 60°C
- (e) Hold 30 min at 60°C
- (f) Sampling and check shades

Remark : No necessary to add salt and alkali in shade correction process

II Information of Dyes Property and Fastness

Basic Condition: a. material: un-mercerized cotton knits
b. chemicals: Glauber's salt and soda ash
c. dyeing temperature: (1) regular dyes at 60°C
(2) Lemon ERC & Turquoise Blue ERC at 80°C
d. dyeing depth: regular colors at 3% owf, blacks at 6% o.w.f
e. liquor ratio: 1 to 10

1. Exhaustion and fixation curves

Assessment: initiate dyeing with salt and dyes at room temperature, then heat up to 60°C, then hold 20 minutes. Add alkali and hold 60 minutes for fixation.

2. Reactivity

The classification was made on the basis of an amount of alkali(soda ash) to achieve the maximum strength

pH value	Reactivity group	Classification
<11	I	High reactivity
11-11.5	II	Medium reactivity
>11.5	III	Low reactivity

3. Substantivity

The classification was made on the basis of a dye-uptake before the alkali addition

Dye-uptake	Substantivity group	Classification
>75%	A	High substantivity
40-75%	B	Medium substantivity
<40%	C	Low substantivity

4. Migration

The tested fabric absorbs dyes and salt in primary exhaustion then exchanges together with an undyed fabric in a blank bath with salt 60 g/l only at liquor ratio 1 to 10, then hold 30 minutes, then add 20 g/l soda ash for fixation.

Assessment: compare the original tested fabric after migration with the fabric after migration by means of AATCC grey scale for assessing color change.

5. Wash-off property

Control the pick-up at 100% of dye liquor on the dyed samples based on related dyeing depth. The following process will define the wash-off property of unfixed dyes.

- | | |
|---------------------|-----------------------------------|
| (1) 60°C x 5 minute | Warm rinsing |
| (2) 80°C x 5 minute | Hot rinsing |
| (3) 98°C x 5 minute | Soaping with soaping agent, 3 g/l |
| (4) 80°C x 5 minute | Hot rinsing |
| (5) 60°C x 5 minute | Warm rinsing |

$$\text{Assessment: } \frac{\text{Amount removed at step (1) and (2)}}{\text{Total amount removed}} \times 100\% = \text{Wash-Off}(\%)$$

Classification

>96% : 5	71 - 75% : 2-3
91 - 96% : 4-5	60 - 70% : 2
86 - 90% : 4	61 - 65% : 1-2
81 - 85% : 3-4	60%< : 1
76 - 80% : 3	

6. Solubility

(a) Dyes solubility

A predetermined amount of dyes dissolved in warm water at 50°C for 5 minutes, and then filtered with filter paper to define the solubility.

(b) Dyes solubility under presence of 50 g/l Glauber's salt

A predetermined amount of dyes and 50 g/l Glauber's salt dissolved in warm water at 50°C for 5 minutes, then filtered with filter paper to define the solubility.

7. Stripping (liquor ratio 1 to 10)

(1) Alkali treatment

Caustic soda (38°B'e)	20 ml/l
Temp. x Time	80°C x 30 minutes
Warm rinsing ⇒ cold rinsing	

(2) Reduction treatment

Caustic soda (38°B'e)	20 ml/l
Sodium hydrosulfite	5 g /l
Temp. x Time	80°C x 30 minutes
Warm rinsing ⇒ cold rinsing	

(3) Combined reduction and oxidation treatment

After the treatment (2), the fabrics are treatment in the following chlorine bath:

Sodium hypochlorite (available chlorine 12%)	8 g /l
Temp. x Time	30°C x 30 minutes
Warm rinsing ⇒ cold rinsing	

(4) Combined oxidation and reduction treatment

Sodium hypochlorite (available chlorine 12%)	8 g /l
Liquor ratio	1 : 10
Temp. x Time	30°C x 30 minutes
Warm rinsing ⇒ cold rinsing	

After the treatment and cold rinsing, then further treated under the condition(2).

Assessment: Assessing the stripping degree by means of the grey scale compared with the un-dyed fabric.

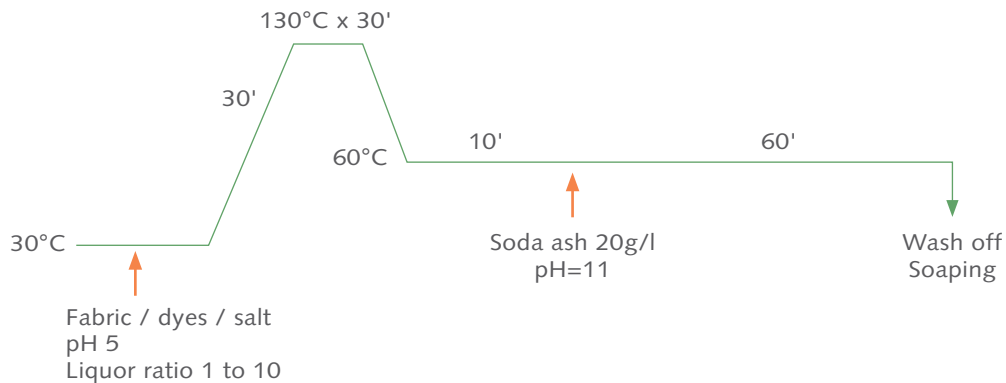
- Classification: (1)  unsuitable (below grade 2)
(2)  moderate (between grade 2-3 to 3)
(3)  suitable (above grade 3-4)

8. Suitability to high temperature in one bath dyeing

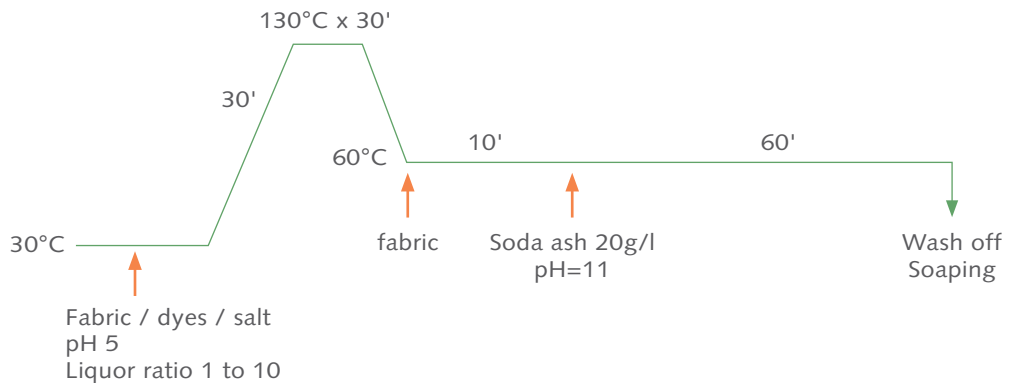
(a) Stability to dyed fabric

The shade change of dyed fabric will be subject to the below high temperature to assess the suitability for polyester/cotton blends dyeing.

Assessment: compare the shade change between normal dyeing with below process by means of grey scale.



(b) Stability to dye liquor



9. Color Constancy

Shade under TL84 and Tungsten lamps were compared with the standard light source(D65) and shade change assessed.

Symbols description of different shade tendency

Y : Yellower R : Redder B : Bluer G : Greener

10. Effect of metallic ion in dye bath

(a) Fe^{2+} : 20 ppm by ferrous chloride (FeCl_2)

(b) Cu^{2+} : 20 ppm by copper sulfate (CuSO_4)

11. Shade variation by fixing agent

(a) Polyethylene polyamide 2% o.w.f 60°C x 20 minutes

(b) Quaternary Ammonium salt 2% o.w.f 60°C x 20 minutes

Assessment: compared those test specimens before and after fixing treatment by means of grey scale for color change.

12. Shade variation after resin finishing

(1) Recipe of resin treatment:

A. Melamine resin		B. Glyoxal resin	
Resin M-3	40 g	Resin GF-4000	40 g
Catalyst MRK	8 g	Catalyst G	8 g
Water	X g	Water	X g
Total		Total	1000 g

(2) Process: resin pick-up 70% ➡ drying at 100°C x 3 minutes ➡ curing at 160°C x 3 minutes

13. Dischargeability

The dyed fabrics will be proceeded by process below:

(1) Recipe of discharge paste:

	Neutral	Alkaline
Rongalit C	200	200
Soda ash	–	50
Hygroscopic Agent	X	X
Thickener	Y	Y
Water	Z	Z
Total	1000 parts	1000 parts

(2) Process: printing ➡ drying at 100°C ➡ steaming at 102°C for 5-8 minutes ➡ rinsing and soaping.

14. Dependency on temperature (fixation time for 60 minutes after alkali addition)

Temperature (°C)	40	50	60	70	80	90	98
Regular dyes	✓	✓	✓	✓	✓		
Lemon/Turquoise dyes			✓	✓	✓	✓	✓

15. Dependency on Glauber's salt (fixation time for 60 minutes after alkali addition)

Dyeing depth	Refer the data of each Everzol ERC dyes					
Dosage (g /l)	30	40	50	60	70	80

16. Dependency on liquor ratio (standard control at 1:10)

Dyeing depth	Refer the data of each Everzol ERC dyes										
Liquor ratio at	5	8	10	12	15	20	30	40	60	80	120

17. Dependency on alkali (standard control by 20 g/l soda ash)

Dyeing depth	Refer the data of each Everzol ERC dyes					
Alkali type & dosage	(a)	(b)	(c)	(d)	(e)	(f)
Soda ash (g/l)	5	10	15	20	5	5
Caustic soda ((38°B'e) ml/l	-	-	-	-	1	2

18. Color Fastness

1. Washing	ISO 105 C10 B ISO 105 C10 C ISO 105 C10 D ISO 105 C06-C15 ISO 105 C06-C25 Marks & Spencer C4A
2. Water	ISO 105 E01
3. Hot water	ISO 105 E08
4. Sea water	ISO 105 E02
5. Post Mercerization	ISO 105 X04
6. Acid hydrolysis	Immerse 1% acetic acid solution for 30 min, then keep 38°C x 6 hours
7. Perspiration (acid / alkali)	ISO 105 E04
8. Chlorinated water (20ppm)	ISO 105 E03
9. Peroxide bleaching	ISO 105 N02
10. Rubbing (dry & wet)	ISO 105 X12
11. Detergent light	Marks & Spencer C9A
12. Multi-c ycle laundry	ISO 105 C09
13. Saliva	DIN 53160
14. Hot pressing	ISO 105 X11
15. Light	ISO 105 B02 & AATCC 16-3
16. Perspiration light	ISO 105 B07 & AATCC 125
17. Nitrogen oxides	ISO 105-G01 (1 cycle)

Everzol

染色需求 Positioning and Segmentation	主要三原色 Main Combination	輔助色 Supplementary
高日光淺中色 Excellent Lightfastness Pale-Medium Shade	Amber ERC Light Red ERC Blue ERC	Yellow ERC
無光變性淺中色 Non-Photochromic Pale-Medium Shade	Amber ERC Coral ERC Blue ERC	Pink ERC Rose ERC
經濟中三原色 Economic Medium Shade	Yellow ERC Rose ERC Navy ERC	Blue ERC
中深三原色 Medium-Deep Shade	Yellow ERC Cardinal ERC Bordeaux ERC Dark Blue ERC	Burnt Orange ERC
深三原色 Deep Shade	Deep Orange ERC Cardinal ERC Bordeaux ERC Dark Blue ERC	Burnt Orange ERC Marine ERC
亮緋紅、大紅色 Brilliant Deep Scarlet and Deep Red Shade	Carrot ERC Saffron ERC Cardinal ERC	Pink ERC
藏青色系 Deep Navy Shade	Bordeaux ERC Marine ERC	Yellow ERC Rose ERC Navy ERC
翠綠色系 Jade Green Shade	Lemon ERC Sky Blue ERC Turquoise Blue ERC	Blue ERC
黑色系 Black Shade	Deep Orange ERC Cardinal ERC Black ERC Jet Black ERC	Yellow ERC Burnt Orange ERC Bordeaux ERC Marine ERC

Everzol ERC Dyes

頁 Page	染料 Dyes	1%	3%	染料特性	Characteristics
1	Lemon ERC			1. 極佳的染深性 2. 與 Everzol Turq. Blue ERC 配色時，染色溫度為 80° C	1. Excellent build-up 2. Optimum dyeing temperature at 80°C while combined with Everzol Turq. Blue ERC
3	Amber ERC			1. 無光變色性之黃色染料 2. 極佳的均染性及染色再現性	1. Gloden yellow dyes with non-photochromic 2. Good leveling and reproducibility properties
5	Yellow ERC			1. 易洗、節能、環保、高牢度 2. 優異染深性 3. 優異日光牢度 4. 中深色三原色	1. Easy wash-off & saving energy & eco-friendly 2. Unique build-up 3. Unique lightfastness 4. Main trichromatic for medium/dark shade
7	Carrot ERC			1. 適合艷橙、大紅色流行色 2. 高彩度、高染深性 3. 高日光、高汗日光、高耐氯水牢度	1. Suitable for very brilliant orange, scarlet fashionable colors 2. High chromatic, good build-up 3. Exceptional lightfastness, perspiration light fastness, chlorinated water fastness
9	Saffron ERC			1. 適合於鮮豔緋紅、大紅流行色系 2. 高彩度、高染深性、洗淨性佳 3. 水洗牢度優異	1. Suitable for very brilliant scarlet, red fashionable colors 2. High chromatic, good build-up, easy wash-off 3. Unique washing fastness
11	Coral ERC			1. 紅色三原色，配色色域廣 2. 與 Amber ERC、Blue ERC 染極淺色時，均染性再現性佳	1. Tinctorial red, broad color yield for color match 2. Compatible with Amber ERC, Blue ERC at very pale shade dyeing with outstanding leveling and reproducibility
13	Pink ERC			1. 鮮豔的紅色 2. 紅色淺三原色，配色色域廣 3. 良好的日光牢度	1. Brilliant red dyes 2. Primary light red, broad color range for color match 3. Good lightfastness
15	Light Red ERC			1. 適合於淺 / 中色系 2. 優異日光牢度及耐氯水牢度 3. 應用性廣	1. Suitable for light-medium shades 2. Unique lightfastness/chlorinated water fastness 3. Suitable for exhaust and CPB dyeing
17	Rose ERC			1. 紅色三原色，配色色域廣 2. 極佳之耐氯水牢度 3. 耐多次水洗牢度佳	1. Primary red, broad color gamut for color match 2. Excellent chlorinated water fastness 3. Unique UK-TO washing fastness
19	Burnt Orange ERC			1. 極佳的染深性 2. 優異的日光牢度及汗日光牢度 3. 適合防拔染	1. Excellent build-up 2. Outstanding lightfastness and perspiration lightfastness 3. Suitable for resist, discharge printing
21	Deep Orange ERC			1. 超經濟、超染深 2. 高日光、汗日光複合牢度 3. 優異耐氯水牢度	1. Extremely high buid-up & cost-effective 2. Excellent lightfastness & perspiration lightfastness 3. Excellent chlorinated water fastness

Everzol ERC Dyes

頁 Page	染料 Dyes	1%	3%	染料特性	Characteristics
23	Cardinal ERC			1. 易洗、節能、環保、高牢度 2. 色味偏黃之大紅色，優異染深性 3. 優異日光牢度及優異耐氯水牢度及水洗牢度 4. 適用於淺色至深色	1. Easy wash-off & saving energy & eco-friendly 2. Yellowish red dyes, unique build-up 3. Excellent fastness on light & chlorine & washing 4. Suitable for pale to dark shade
25	Bordeaux ERC			1. 易洗、節能、環保、高牢度 2. 色味偏藍之大紅色，優異染深性 3. 優異耐氯漂牢度及水洗牢度 4. 中深三原色	1. Easy wash-off & saving energy & eco-friendly 2. Bluish red dyes with unique build-up 3. Excellent fastness on chlorine & washing 4. Trichromatic for medium / dark shade
27	Turquoise Blue ERC			1. 適合艷綠色配色組合 2. 與 Lemon ERC 配色染色時，染色溫度 80° C 3. 鹽類添加推薦使用芒硝	1. Suitable for color match of brilliant green shade 2. Optimum dyeing temperature at 80°C while combined with Lemon ERC 3. Glauber's salt is highly recommended
29	Sky Blue ERC			1. 鮮豔偏綠的寶藍色染料 2. 極佳的日光牢度 3. 與 Lemon ERC、Turquoise Blue ERC 可配出鮮艷的翠綠色 4. N/C 混紡染色，低 Nylon 污染	1. Brilliant greenish royal blue dyes 2. Excellent lightfastness 3. Compatible with Lemon ERC, Turquoise Blue ERC for matching brilliant jade green shade 4. Low nylon cross-dyeing for Nylon/Cotton blends dyeing
31	Blue ERC			1. 易洗、節能、環保、高牢度 2. 藍色三原色，配色色域廣 3. 優異水洗牢度及日光牢度	1. Easy wash-off & saving energy & eco-friendly 2. Tinctorial blue, wide gamut for color match 3. Unique fastness on washing & light
33	Dark Blue ERC			1. 易洗、節能、環保、高牢度 2. 優異染深性之深藍色染料 3. 優異水洗牢度及日光牢度 4. 中深色三原色，均染性極佳	1. Easy wash-off & saving energy & eco-friendly 2. Unique build-up for dark blue shade 3. Outstanding fastness on washing & light 4. Main trichromatic for medium/deep shade & excellent levelness
35	Navy ERC			1. 丈青色三原色，配色色域廣 2. 極佳之耐氯水牢度 3. 耐多次水洗牢度佳 4. 均染性及再現性佳	1. Primary navy, broad color range for color match 2. Excellent chlorinated water fastness 3. Unique UK-TO washing fastness 4. Excellent leveling and reproducibility
37	Marine ERC			1. 易洗、節能、環保、高牢度 2. 優異染深性 3. 優異耐氯水及 UK-TO 牢度 4. 深色三原色	1. Easy wash-off & saving energy & eco-friendly 2. Unique build-up 3. Outstanding fastness on chlorine & UK-TO washing 4. Main trichromatic for deep shade
39	Black ERC			1. 適合防、拔染底色 2. 耐後絲光牢度佳 3. 耐多次水洗牢度佳	1. Suitable for resist, discharge printing 2. Excellent colorfastness to post mercerizing 3. Excellent UK-TO fastness
41	Jet Black ERC			1. 適合於深黑色系 2. 適用於省水節能 Everzol ERC-MTS (70° C) 洗程 3. 可應用於浸染及 CPB 染色	1. Suitable for deep black shades 2. Suitable for Everzol ERC-MTS (70°C) eco wash-off 3. Suitable for exhaust and CPB dyeing processes

註：染色濃度為 1% & 3%，黑色系染色濃度為 3% & 6%

Notes : The dyeing depth at 1% & 3%, Black shades the dyeing depth at 3% & 6%

棉用反應性染料 REACTIVE DYES

Everzol[®] ERC Solution

Eco Resource Conservation

- 色域完整
- 省水、省能源
- 環保、易洗淨
- 高均染、重現性
- 高牢度表現
- 完整解決方案
- Full Gamut by Everzol ERC Dyes
- Save Water and Energy
- Eco-Friendly and Easy Wash-Off
- Unique Levelness and R.F.T.
- Outstanding Fastness Performance
- Sustainability Solution



水 Water

-70%



蒸汽 Steam

-80%



電 Electricity

-67%



時間 Time

-47%



碳排放 CO₂

-80%



**Go Green
Go Fast**

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Tel: +886-2-27066006
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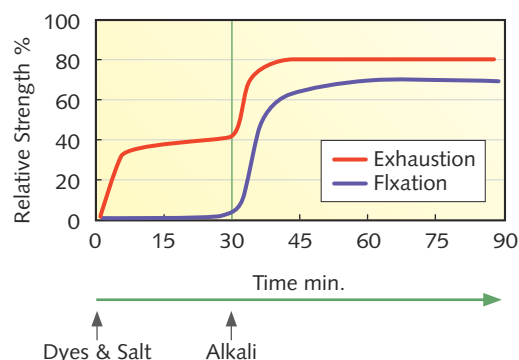



**Everlight
Chemical**

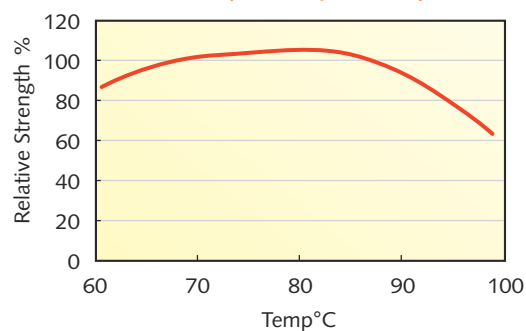
Dyeing Properties 特性

Characteristics		Evaluation
Reactivity 反應性		II
Substantivity 親和性		B
Migration 移染性		4-5
Washing Off 洗淨性		3-4
Solubility g/l 溶解度	50°C Water	100
	50°C Water + 50 g/l Salt	100
High Temp. Dyeing 高溫染色性	Dyed Fabric 色布	2-3R
	Dye Liquor 染浴	3-4R
Color Constancy 色恆性	Tungsten Lamp	G
	TL 84 Lamp	G
Metallic Ions 金屬離子	Fe ²⁺ 鐵	4-5
	Cu ²⁺ 銅	2-3
Resin Finishing 樹脂加工	Melamine 三聚氰胺	5
	Glyoxal 乙二醛	5
Shade Change after Fixing Treatment 固色後變色性	Polyamine 聚胺類	4-5
	Quaternary Ammonium Salt 四級銨鹽	4-5
Stripping 脫色性	(1)	X
	(2)	X
	(3)	△
	(4)	○
Discharge Ability 拔染性	Neutral 中性	4-5
	Alkaline 鹼性	4-5

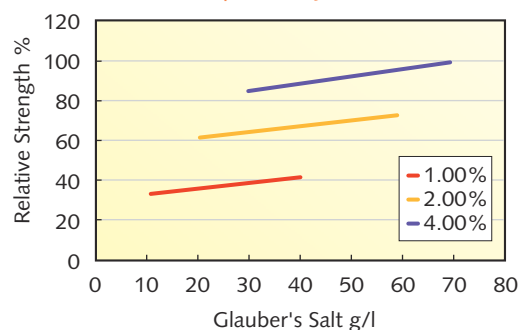
吸盡固著曲線 Exhaustion & Fixation



溫度依存性 Dependency on Temperature



芒硝依存性 Dependency on Glauber's Salt



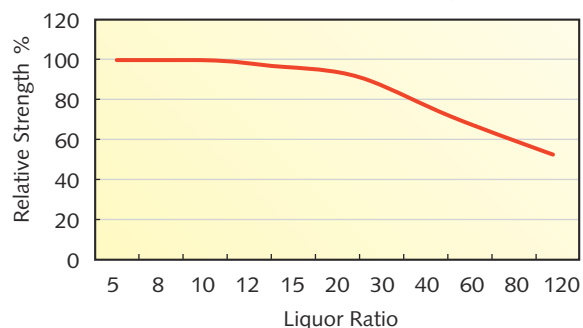
Fastness before Fixing Treatment 固色前牢度表

Evaluation		Depth of Shade 染色濃度	Color Change 變退色	Staining 污染						Evaluation		Depth of Shade 染色濃度	Grade 級數	
Fastness				Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛	Fastness				
Washing 耐水洗	ISO C10 B	3.0%	5	5	5	5	5	5	5	Chlorinated Water 耐氯水 (20ppm)		3.0%	3	
	ISO C10 C		5	5	5	5	5	5	5	Peroxide Bleaching 雙氧水漂白			2-3	
	ISO C10 D		5	5	4	5	5	5	3-4	Rubbing 耐摩擦	Dry 乾	3.0%	5	
	ISO C06-C1S		4-5	5	4-5	5	5	5	5		Wet 濕		3-4	
	ISO C06-C2S		4-5	5	4-5	5	5	5	5	M&S C9A 擬濕日光 (Detergent Light)		0.3%	4-5	
	M&S C4A		4-5	5	4-5	5	5	5	5			3.0%	4-5	
Water 耐水		3.0%	4-5	5	5	5	5	5	5	Multi-Washing 耐多次水洗		3.0%	3	
Hot Water 耐熱水		3.0%	4-5	5	5	5	5	5	5	Saliva (DIN) 耐唾液		3.0%	5	
Sea Water 耐海水		3.0%	4-5	5	4-5	5	5	5	5	Hot-Pressing 耐熱壓	Dry 乾	Immediately	3.0%	5
Post Mercerizing 後絲光		3.0%	4-5		4-5						4 hours after	5		
Acid Hydrolysis 酸水解		3.0%	4-5	5	4-5	3-4	5	5	3-4		Wet 濕	Immediately	3.0%	5
Perspiration 耐汗	Acid 酸	3.0%	5	5	5	5	5	5	5			4 hours after		5
	Alkali 鹼		5	5	5	5	5	5	5	Nitrogen Oxides 氣體變褪色		3.0%	5	

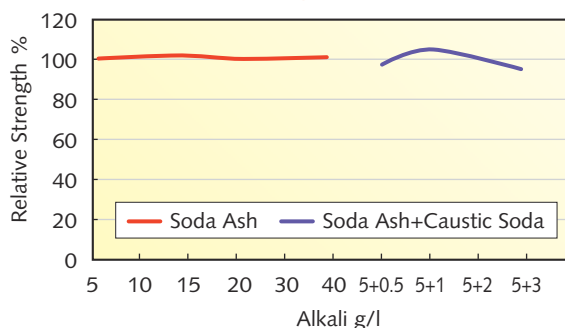
Light Fastness 日光堅牢度

Evaluation	Dyeing Depth	Color Change	ISO Blue	Evaluation		Dyeing Depth	Color Change	Grade ISO
Fastness	染色濃度	20 AFU	Scale	Fastness		染色濃度	20 AFU	Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.1%	*	*	Perspiration Light 耐汗日光	Acid	3.0%	2-3	5-6
	0.3%	4	5		Alkali		4	3-4
	1.0%	4-5	6					
	3.0%	4-5	>6					
	6.0%	4-5	>6					

浴比依存性 Dependency on Liquor Ratio



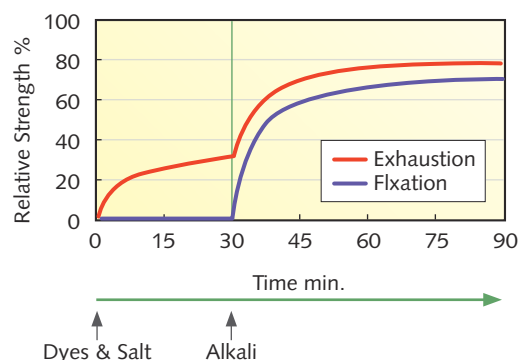
鹼依存性 Dependency on Alkali



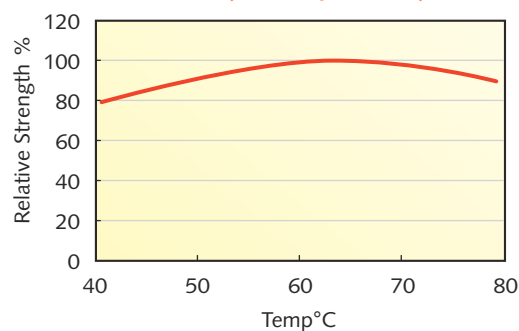
Dyeing Properties 特性

Characteristics		Evaluation
Reactivity 反應性		II
Substantivity 親和性		C
Migration 移染性		3-4
Washing Off 洗淨性		4-5
Solubility g/l 溶解度	50°C Water	100
	50°C Water + 50 g/l Salt	100
High Temp. Dyeing 高溫染色性	Dyed Fabric 色布	3
	Dye Liquor 染浴	3
Color Constancy 色恆性	Tungsten Lamp	B(G)
	TL 84 Lamp	G(Y)
Metallic Ions 金屬離子	Fe ²⁺ 鐵	4-5
	Cu ²⁺ 銅	3-4
Resin Finishing 樹脂加工	Melamine 三聚氰胺	4-5
	Glyoxal 乙二醛	4-5
Shade Change after Fixing Treatment 固色後變色性	Polyamine 聚胺類	4-5
	Quaternary Ammonium Salt 四級銨鹽	4-5
Stripping 脫色性	(1)	X
	(2)	X
	(3)	X
	(4)	△
Discharge Ability 拔染性	Neutral 中性	3-4
	Alkaline 鹼性	1-2

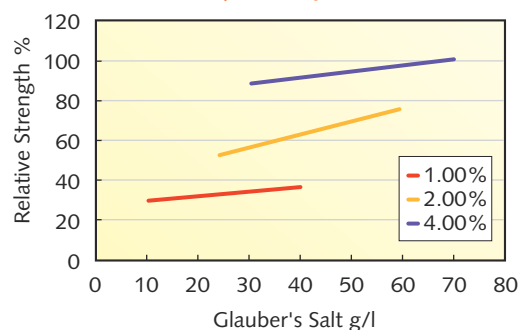
吸盡固著曲線 Exhaustion & Fixation



溫度依存性 Dependency on Temperature



芒硝依存性 Dependency on Glauber's Salt



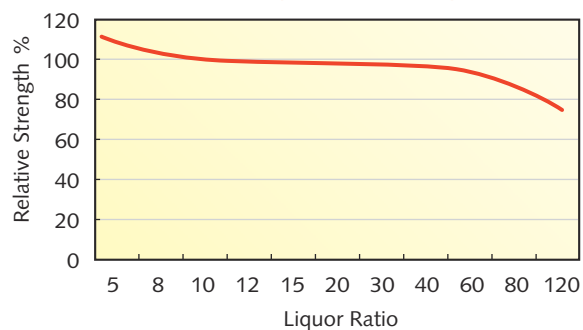
Fastness before Fixing Treatment 固色前牢度表

Evaluation		Depth of Shade 染色濃度	Color Change 變退色	Staining 污染						Evaluation			Depth of Shade 染色濃度	Grade 級數
Fastness				Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛	Fastness				
Washing 耐水洗	ISO C10 B	3.0%	4-5	4-5	4	4-5	4-5	4-5	4-5	Chlorinated Water 耐氯水 (20ppm)			3.0%	4-5
	ISO C10 C		4-5	4-5	4	4-5	4-5	4-5	4-5	Peroxide Bleaching 雙氧水漂白			4-5	
	ISO C10 D		4-5	4-5	3-4	4-5	4-5	4-5	4	Rubbing 耐摩擦	Dry 乾	3.0%	5	
	ISO C06-C1S		4-5	4-5	3-4	4-5	4-5	4-5	4-5		Wet 濕		2-3	
	ISO C06-C2S		4-5	4-5	4	4-5	4-5	4-5	4-5	M&S C9A 擬濕日光 (Detergent Light)			0.3%	4-5
	M&S C4A		4-5	4-5	4	4-5	4-5	4-5	4-5				3.0%	4-5
Water 耐水		3.0%	4-5	5	4-5	4-5	5	5	5	Multi-Washing 耐多次水洗			3.0%	4-5
Hot Water 耐熱水		3.0%	4-5	5	5	4-5	5	5	5	Saliva (DIN) 耐唾液			3.0%	4-5
Sea Water 耐海水		3.0%	4-5	5	5	5	5	5	5	Hot-Pressing 耐熱壓	Dry 乾	Immediately	3.0%	4
Post Mercerizing 後絲光		3.0%	2-3		4				4 hours after			4-5		
Acid Hydrolysis 酸水解		3.0%	4-5	5	4-5	4-5	5	5	5		Wet 濕	Immediately	3.0%	3-4
Perspiration 耐汗	Acid 酸	3.0%	4-5	5	4-5	5	5	5	5			4 hours after		4-5
	Alkali 鹼		4-5	5	4	5	5	5	5	Nitrogen Oxides 氣體變褪色			3.0%	4-5

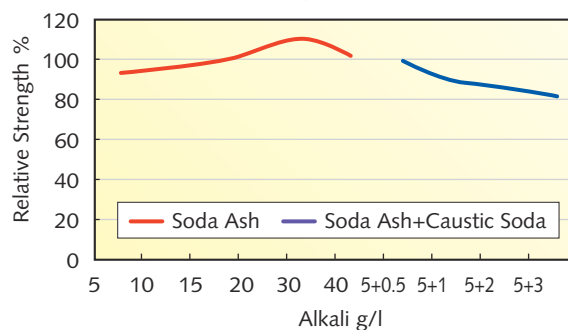
Light Fastness 日光堅牢度

Evaluation		Dyeing Depth 染色濃度	Color Change 20 AFU	ISO Blue Scale	Evaluation		Dying Depth 染色濃度	Color Change 20 AFU	Grade ISO Blue Scale
Fastness					Fastness				
Artificial Light (Xenon-Arc Lamp) 耐日光	0.1%	4-5	5-6	Perspiration Light 耐汗日光	Acid	3.0%	4-5	>6	
	0.3%	4-5	6		Alkali		4-5	>6	
	1.0%	4-5	>6						
	3.0%	4-5	>6						
	6.0%	4-5	>6						

浴比依存性 Dependency on Liquor Ratio



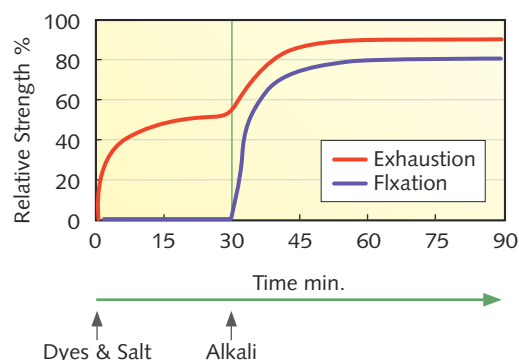
鹼依存性 Dependency on Alkali



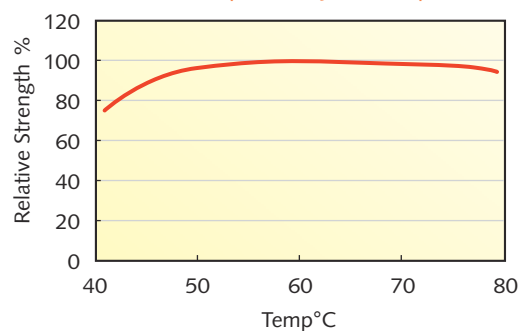
Dyeing Properties 特性

Characteristics		Evaluation
Reactivity 反應性		II
Substantivity 親和性		B
Migration 移染性		4
Washing Off 洗淨性		4
Solubility g/l 溶解度	50°C Water	100
	50°C Water + 50 g/l Salt	100
High Temp. Dyeing 高溫染色性	Dyed Fabric 色布	3-4
	Dye Liquor 染浴	4
Color Constancy 色恆性	Tungsten Lamp	R
	TL 84 Lamp	R
Metallic Ions 金屬離子	Fe ²⁺ 鐵	4-5
	Cu ²⁺ 銅	4
Resin Finishing 樹脂加工	Melamine 三聚氰胺	5
	Glyoxal 乙二醛	5
Shade Change after Fixing Treatment 固色後變色性	Polyamine 聚胺類	4-5
	Quaternary Ammonium Salt 四級銨鹽	4-5
Stripping 脫色性	(1)	X
	(2)	X
	(3)	△
	(4)	△
Discharge Ability 拔染性	Neutral 中性	3
	Alkaline 鹼性	2-3

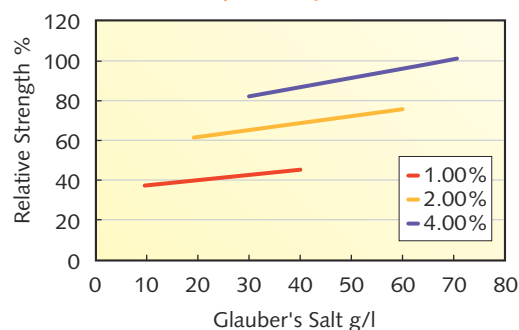
吸盡固著曲線 Exhaustion & Fixation



溫度依存性 Dependency on Temperature



芒硝依存性 Dependency on Glauber's Salt



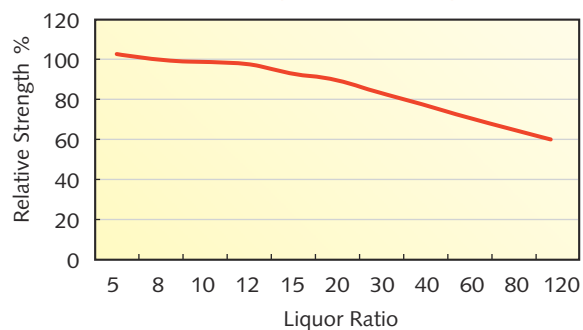
Fastness before Fixing Treatment 固色前牢度表

Evaluation		Depth of Shade 染色濃度	Color Change 變退色	Staining 污染						Evaluation			Depth of Shade 染色濃度	Grade 級數
Fastness				Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛	Fastness				
Washing 耐水洗	ISO C10 B	3.0%	4-5	5	4-5	5	5	5	5	Chlorinated Water 耐氯水 (20ppm)			3.0%	2-3
	ISO C10 C		4-5	5	4-5	5	5	5	5	Peroxide Bleaching 雙氧水漂白				4-5
	ISO C10 D		4-5	5	4	5	5	5	4	Rubbing 耐摩擦	Dry 乾	3.0%	5	
	ISO C06-C1S		4-5	5	4-5	5	5	5	5		Wet 濕		3-4	
	ISO C06-C2S		4-5	5	4-5	5	5	5	5	M&S C9A 擬濕日光 (Detergent Light)			0.3%	5
	M&S C4A		4-5	5	4-5	5	5	5	5				3.0%	5
Water 耐水		3.0%	4-5	5	5	5	5	5	5	Multi-Washing 耐多次水洗			3.0%	4-5
Hot Water 耐熱水		3.0%	4-5	5	5	5	5	5	5	Saliva (DIN) 耐唾液			3.0%	5
Sea Water 耐海水		3.0%	4	5	4-5	5	5	5	5	Hot-Pressing 耐熱壓	Dry 乾	Immediately	3.0%	5
Post Mercerizing 後絲光		3.0%	4		4-5							4 hours after		5
Acid Hydrolysis 酸水解		3.0%	4-5	5	4-5	4	5	5	4-5		Wet 濕	Immediately	3.0%	4-5
Perspiration 耐汗	Acid 酸	3.0%	4-5	5	4-5	5	5	5	5			4 hours after		5
	Alkali 鹼		4-5	5	4-5	5	5	5	5	Nitrogen Oxides 氣體變褪色			3.0%	5

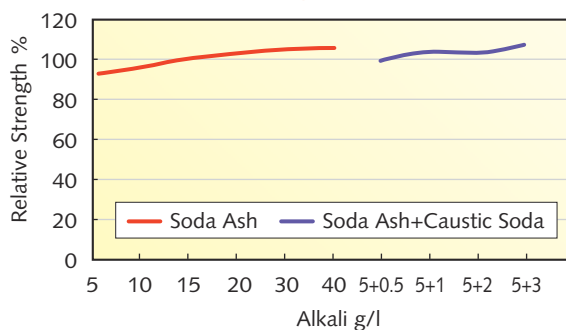
Light Fastness 日光堅牢度

Evaluation		Dyeing Depth 染色濃度	Color Change 20 AFU	ISO Blue Scale	Evaluation		Dying Depth 染色濃度	Color Change 20 AFU	Grade ISO Blue Scale
Fastness					Fastness				
Artificial Light (Xenon-Arc Lamp) 耐日光	0.1%	*	*	Perspiration Light 耐汗日光	Acid	3.0%	4	5-6	
	0.3%	4-5	5-6		Alkali		4	5	
	1.0%	4-5	>6						
	3.0%	4-5	>6						
	6.0%	4-5	>6						

浴比依存性 Dependency on Liquor Ratio



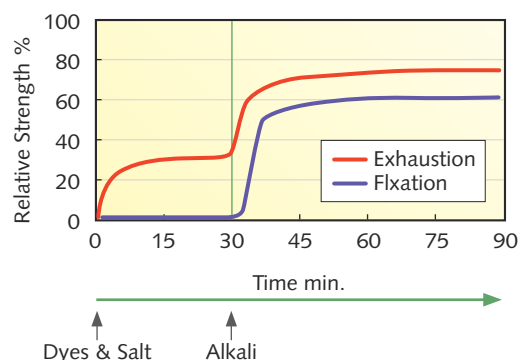
鹼依存性 Dependency on Alkali



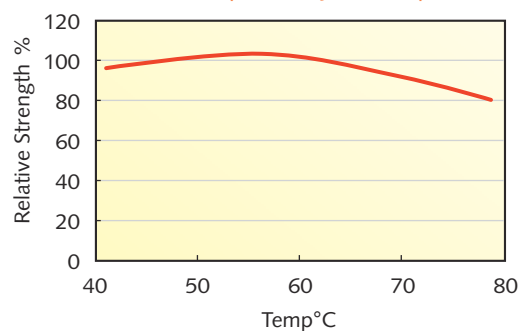
Dyeing Properties 特性

Characteristics		Evaluation
Reactivity 反應性		II
Substantivity 親和性		C
Migration 移染性		4-5
Washing Off 洗淨性		3-4
Solubility g/l 溶解度	50°C Water	100
	50°C Water + 50 g/l Salt	100
High Temp. Dyeing 高溫染色性	Dyed Fabric 色布	4-5
	Dye Liquor 染浴	5
Color Constancy 色恆性	Tungsten Lamp	Y(G)
	TL 84 Lamp	Y(G)
Metallic Ions 金屬離子	Fe ²⁺ 鐵	4-5
	Cu ²⁺ 銅	2
Resin Finishing 樹脂加工	Melamine 三聚氰胺	4
	Glyoxal 乙二醛	4
Shade Change after Fixing Treatment 固色後變色性	Polyamine 聚胺類	4-5
	Quaternary Ammonium Salt 四級銨鹽	4-5
Stripping 脫色性	(1)	X
	(2)	△
	(3)	△
	(4)	△
Discharge Ability 拔染性	Neutral 中性	3
	Alkaline 鹼性	3-4

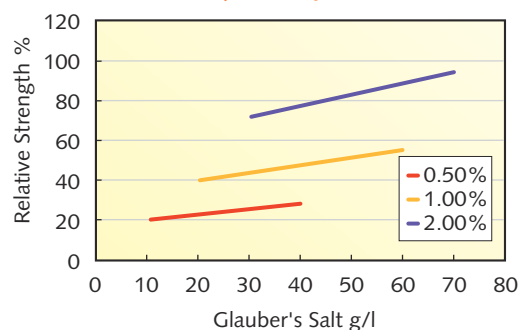
吸盡固著曲線 Exhaustion & Fixation



溫度依存性 Dependency on Temperature



芒硝依存性 Dependency on Glauber's Salt



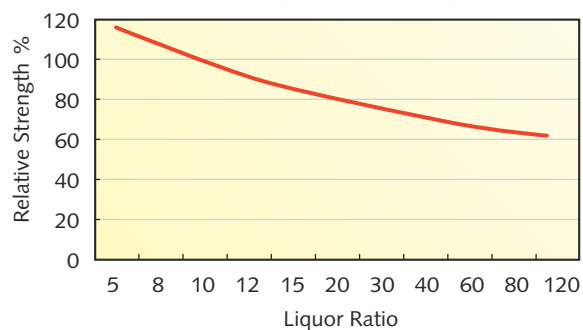
Fastness before Fixing Treatment 固色前牢度表

Evaluation		Depth of Shade 染色濃度	Color Change 變退色	Staining 污染						Evaluation			Depth of Shade 染色濃度	Grade 級數
Fastness				Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛	Fastness				
Washing 耐水洗	ISO C10 B	3.0%	4-5	5	4-5	5	5	5	5	Chlorinated Water 耐氯水 (20ppm)			3.0%	4-5
	ISO C10 C		4-5	5	4-5	5	5	5	5	Peroxide Bleaching 雙氧水漂白				3-4
	ISO C10 D		4-5	5	4	5	5	4-5	4	Rubbing 耐摩擦	Dry 乾	3.0%	4-5	
	ISO C06-C1S		4-5	5	4-5	5	5	5	5		Wet 濕		3	
	ISO C06-C2S		4-5	5	4-5	5	5	5	5	M&S C9A 擬濕日光 (Detergent Light)			0.3%	4-5
	M&S C4A		4-5	5	4-5	5	5	5	5				3.0%	4-5
Water 耐水		3.0%	4-5	5	4	4	5	4	4	Multi-Washing 耐多次水洗			3.0%	4
Hot Water 耐熱水		3.0%	4-5	5	4-5	4-5	5	5	5	Saliva (DIN) 耐唾液			3.0%	4-5
Sea Water 耐海水		3.0%	4-5	5	4-5	5	5	5	5	Hot-Pressing 耐熱壓	Dry 乾	Immediately	3.0%	3-4
Post Mercerizing 後絲光		3.0%	4		3-4						4 hours after	4-5		
Acid Hydrolysis 酸水解		3.0%	4-5	5	3-4	3-4	5	5	4-5		Wet 濕	Immediately	3.0%	3-4
Perspiration 耐汗	Acid 酸	3.0%	4-5	5	4-5	4-5	5	5	5			4 hours after		4-5
	Alkali 鹼		4-5	5	4-5	4-5	5	5	5	Nitrogen Oxides 氣體變褪色			3.0%	5

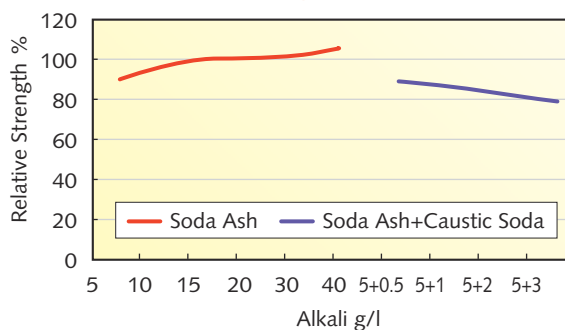
Light Fastness 日光堅牢度

Evaluation	Dyeing Depth	Color Change	ISO Blue	Evaluation		Dyeing Depth	Color Change	Grade ISO
Fastness	染色濃度	20 AFU	Scale	Fastness		染色濃度	20 AFU	Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.1%	3	4-5	Perspiration Light 耐汗日光	Acid	3.0%	4	5-6
	0.3%	3-4	4-5		Alkali		4	5-6
	1.0%	4	5					
	3.0%	4-5	5-6					
	6.0%	4-5	6					

浴比依存性 Dependency on Liquor Ratio



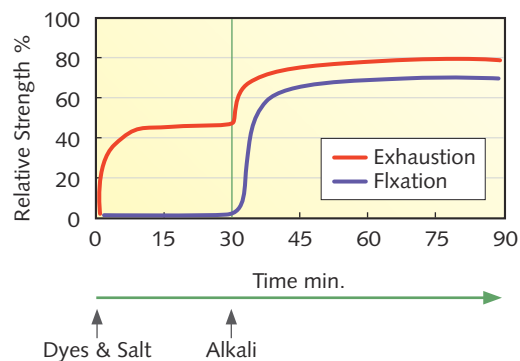
鹼依存性 Dependency on Alkali



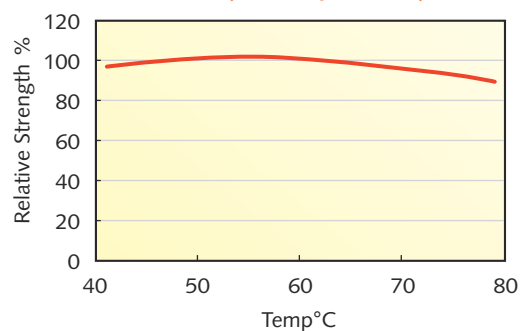
Dyeing Properties 特性

Characteristics		Evaluation
Reactivity 反應性		II
Substantivity 親和性		B
Migration 移染性		4
Washing Off 洗淨性		3
Solubility g/l 溶解度	50°C Water	100
	50°C Water + 50 g/l Salt	90
High Temp. Dyeing 高溫染色性	Dyed Fabric 色布	4-5
	Dye Liquor 染浴	4-5
Color Constancy 色恆性	Tungsten Lamp	Y(G)
	TL 84 Lamp	Y(G)
Metallic Ions 金屬離子	Fe ²⁺ 鐵	5
	Cu ²⁺ 銅	3
Resin Finishing 樹脂加工	Melamine 三聚氰胺	4
	Glyoxal 乙二醛	4-5
Shade Change after Fixing Treatment 固色後變色性	Polyamine 聚胺類	4
	Quaternary Ammonium Salt 四級銨鹽	4-5
Stripping 脫色性	(1)	X
	(2)	○
	(3)	○
	(4)	△
Discharge Ability 拔染性	Neutral 中性	2-3
	Alkaline 鹼性	2-3

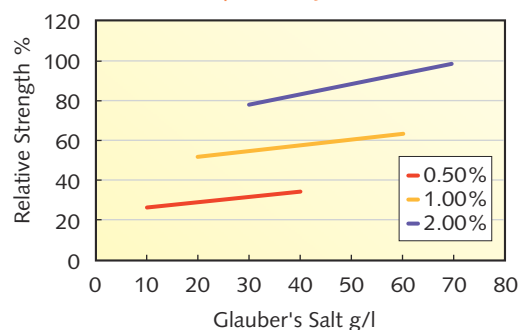
吸盡固著曲線 Exhaustion & Fixation



溫度依存性 Dependency on Temperature



芒硝依存性 Dependency on Glauber's Salt



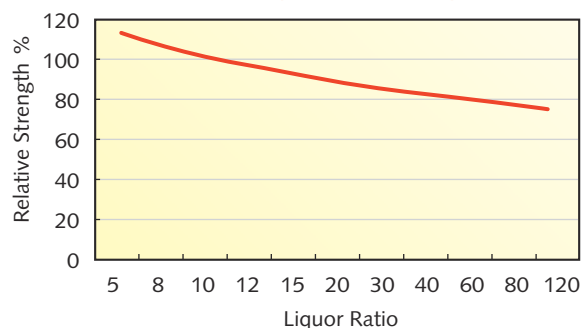
Fastness before Fixing Treatment 固色前牢度表

Evaluation		Depth of shade 染色濃度	Color change 變退色	Staining 污染						Evaluation		Depth of shade 染色濃度	Grade 級數	
Fastness				Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛	Fastness				
Washing 耐水洗	ISO C10 B	3.0%	4-5	5	4-5	5	5	4-5	5	Chlorinated Water 耐氯水 (20ppm)		3.0%	4	
	ISO C10 C		4-5	5	4-5	5	5	4-5	5	Peroxide Bleaching 雙氧水漂白			4-5	
	ISO C10 D		4	5	4	5	5	5	4	Rubbing 耐摩擦	Dry 乾	3.0%	4-5	
	ISO C06-C1S		4-5	5	4	5	5	4-5	5		Wet 濕		2-3	
	ISO C06-C2S		4-5	5	4-5	5	5	5	5	M&S C9A 擬濕日光 (Detergent Light)		0.3%	4-5	
	M&S C4A		4-5	5	5	5	5	5	5			3.0%	4-5	
Water 耐水		3.0%	4-5	5	5	4-5	5	5	5	Multi-Washing 耐多次水洗		3.0%	4-5	
Hot Water 耐熱水		3.0%	4-5	5	5	5	5	5	5	Saliva (DIN) 耐唾液		3.0%	5	
Sea Water 耐海水		3.0%	4-5	5	5	5	5	5	5	Hot-Pressing 耐熱壓	Dry 乾	Immediately	3.0%	2-3
Post Mercerizing 後絲光		3.0%	4		3						4 hours after	4-5		
Acid Hydrolysis 酸水解		3.0%	4-5	5	3-4	4-5	5	4-5	4-5		Wet 濕	Immediately	3.0%	4
Perspiration 耐汗	Acid 酸	3.0%	4-5	5	5	5	5	5	5			4 hours after		4-5
	Alkali 鹼		4-5	4-5	4-5	4-5	4-5	4	4	Nitrogen Oxides 氣體變褪色		3.0%	5	

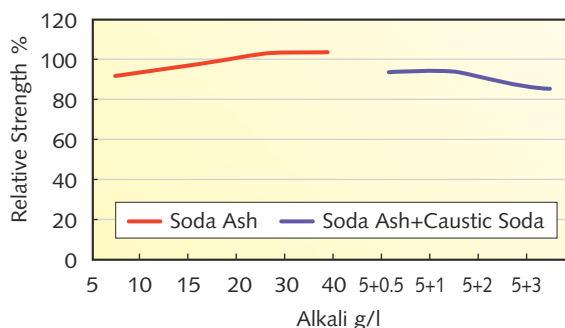
Light Fastness 日光堅牢度

Evaluation	Dyeing Depth	Color Change	ISO Blue	Evaluation		Dying Depth	Color Change	Grade ISO
Fastness	染色濃度	20 AFU	Scale	Fastness		染色濃度	20 AFU	Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.1%	2	3-4	Perspiration Light 耐汗日光	Acid	3.0%	3	5
	0.3%	2-3	4		Alkali		3-4	4-5
	1.0%	3	4-5					
	3.0%	3-4	5					
	6.0%	4	5-6					

浴比依存性 Dependency on Liquor Ratio

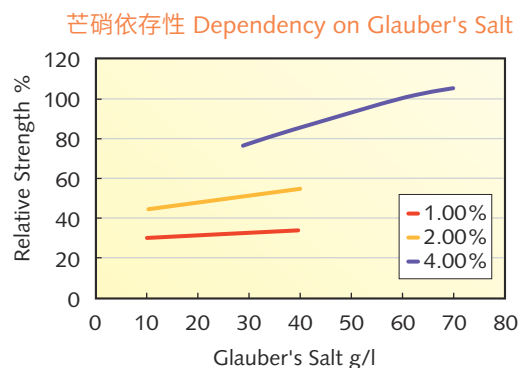
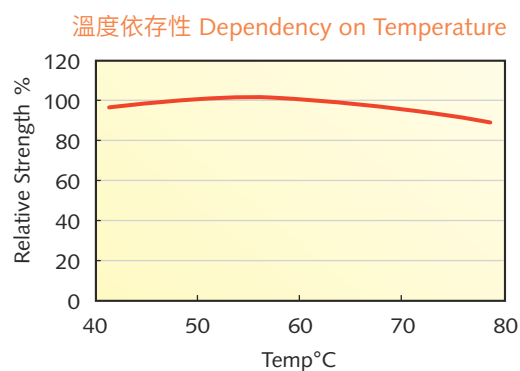
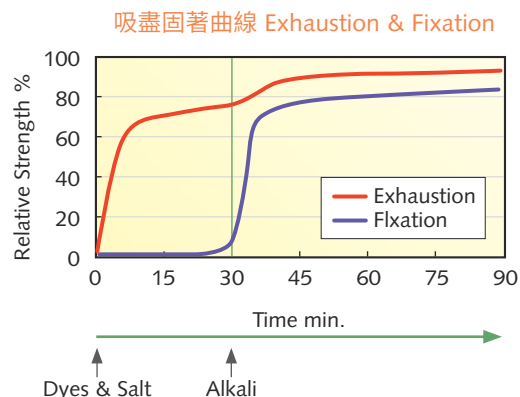


鹼依存性 Dependency on Alkali



Dyeing Properties 特性

Characteristics		Evaluation
Reactivity 反應性		II
Substantivity 親和性		A
Migration 移染性		4
Washing Off 洗淨性		3
Solubility g/l 溶解度	50°C Water	100
	50°C Water + 50 g/l Salt	100
High Temp. Dyeing 高溫染色性	Dyed Fabric 色布	3-4
	Dye Liquor 染浴	4
Color Constancy 色恆性	Tungsten Lamp	R(Y)
	TL 84 Lamp	Y
Metallic Ions 金屬離子	Fe ²⁺ 鐵	4-5
	Cu ²⁺ 銅	1
Resin Finishing 樹脂加工	Melamine 三聚氰胺	4-5
	Glyoxal 乙二醛	4
Shade Change after Fixing Treatment 固色後變色性	Polyamine 聚胺類	4-5
	Quaternary Ammonium Salt 四級銨鹽	4-5
Stripping 脫色性	(1)	X
	(2)	○
	(3)	○
	(4)	○
Discharge Ability 拔染性	Neutral 中性	3-4
	Alkaline 鹼性	3



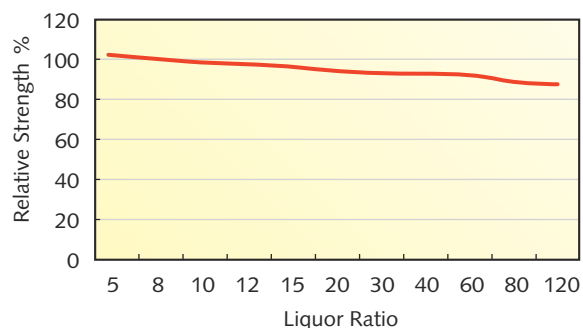
Fastness before Fixing Treatment 固色前牢度表

Evaluation		Depth of Shade 染色濃度	Color Change 變退色	Staining 污染						Evaluation			Depth of Shade 染色濃度	Grade 級數
Fastness				Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛	Fastness				
Washing 耐水洗	ISO C10 B	1.0%	5	5	5	5	5	5	5	Chlorinated Water 耐氯水 (20ppm)			1.0%	4
	ISO C10 C		5	5	5	5	5	5	5	Peroxide Bleaching 雙氧水漂白				4-5
	ISO C10 D		4-5	5	4	5	5	5	5	Rubbing 耐摩擦	Dry 乾	1.0%	5	
	ISO C06-C1S		5	5	5	5	5	5	5		Wet 濕		4	
	ISO C06-C2S		5	5	5	5	5	5	5	M&S C9A 擬濕日光 (Detergent Light)			0.1%	4-5
	M&S C4A		5	5	5	5	5	5	5				1.0%	4-5
Water 耐水		1.0%	5	5	5	5	5	5	5	Multi-Washing 耐多次水洗			1.0%	5
Hot Water 耐熱水		1.0%	5	5	4-5	4-5	5	5	5	Saliva (DIN) 耐唾液			1.0%	4-5
Sea Water 耐海水		1.0%	5	5	5	5	5	5	5	Hot-Pressing 耐熱壓	Dry 乾	Immediately	1.0%	3B
Post Mercerizing 後絲光		1.0%	2-3(B)		4-5							4 hours after		4-5
Acid Hydrolysis 酸水解		1.0%	5	5	4-5	4	5	5	5		Wet 濕	Immediately	1.0%	3-4B
Perspiration 耐汗	Acid 酸	1.0%	5	5	5	5	5	5	5			4 hours after		4-5
	Alkali 鹼		5	5	5	5	5	5	5	5	Nitrogen Oxides 氣體變褪色			1.0%

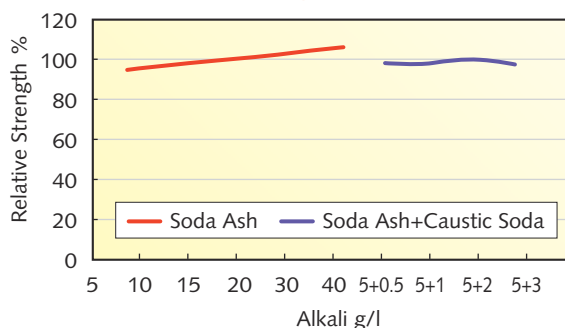
Light Fastness 日光堅牢度

Evaluation	Dyeing Depth	Color Change	ISO Blue	Evaluation		Dying Depth	Color Change	Grade ISO
Fastness	染色濃度	20 AFU	Scale	Fastness		染色濃度	20 AFU	Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.1%	2-3	4	Perspiration Light 耐汗日光	Acid	1.0%	2-3	4-5
	0.3%	3	4-5		Alkali		3	4
	1.0%	3-4	4-5					
	3.0%	4	5-6					
	6.0%	*	*					

浴比依存性 Dependency on Liquor Ratio

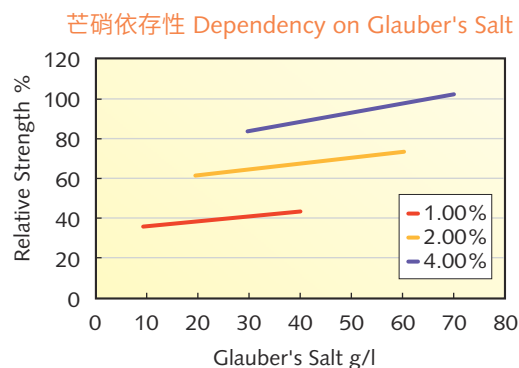
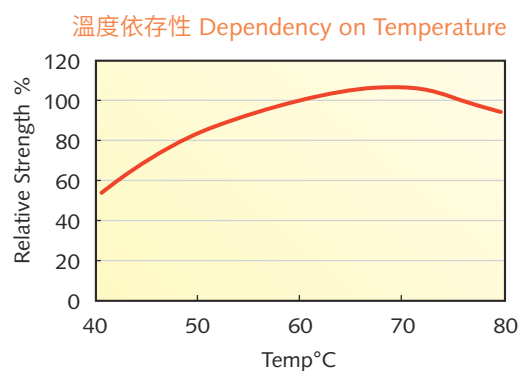
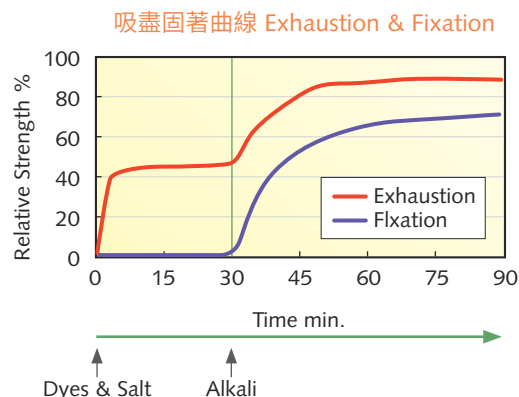


鹼依存性 Dependency on Alkali



Dyeing Properties 特性

Characteristics		Evaluation
Reactivity 反應性		II
Substantivity 親和性		B
Migration 移染性		5
Washing Off 洗淨性		3-4
Solubility g/l 溶解度	50°C Water	50
	50°C Water + 50 g/l Salt	50
High Temp. Dyeing 高溫染色性	Dyed Fabric 色布	3B
	Dye Liquor 染浴	4B
Color Constancy 色恆性	Tungsten Lamp	Y
	TL 84 Lamp	B
Metallic Ions 金屬離子	Fe ²⁺ 鐵	5
	Cu ²⁺ 銅	4
Resin Finishing 樹脂加工	Melamine 三聚氰胺	3B
	Glyoxal 乙二醛	4-5
Shade Change after Fixing Treatment 固色後變色性	Polyamine 聚胺類	4-5
	Quaternary Ammonium Salt 四級銨鹽	5
Stripping 脫色性	(1)	X
	(2)	○
	(3)	○
	(4)	○
Discharge Ability 拔染性	Neutral 中性	1-2
	Alkaline 鹼性	2



Everzol Pink ERC

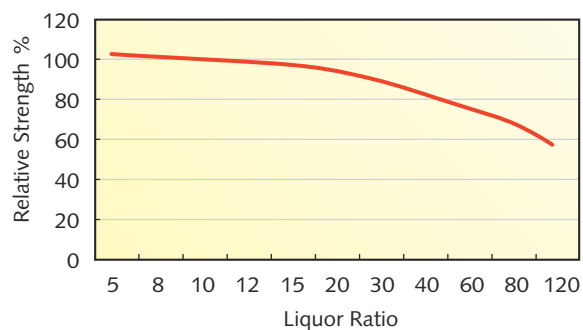
Fastness before Fixing Treatment 固色前牢度表

Evaluation		Depth of Shade 染色濃度	Color Change 變退色	Staining 污染						Evaluation			Depth of Shade 染色濃度	Grade 級數
Fastness				Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛	Fastness				
Washing 耐水洗	ISO C10 B	3.0%	5	5	5	5	5	5	5	Chlorinated Water 耐氯水 (20ppm)			3.0%	4-5
	ISO C10 C		4-5	5	5	5	5	5	5	Peroxide Bleaching 雙氧水漂白				
	ISO C10 D		5	5	5	5	5	5	3	Rubbing 耐摩擦	Dry 乾	3.0%	5	
	ISO C06-C1S		4-5	5	5	5	5	5	5		Wet 濕			
	ISO C06-C2S		4-5	5	5	5	5	5	5	M&S C9A 擬濕日光 (Detergent Light)			0.3%	>4-5
	M&S C4A		4-5	5	5	5	5	5	5				3.0%	>4-5
Water 耐水		3.0%	5	5	4-5	4-5	5	5	5	Multi-Washing 耐多次水洗			3.0%	4-5
Hot Water 耐熱水		3.0%	4-5	4-5	5	5	5	5	5	Saliva (DIN) 耐唾液			3.0%	4-5
Sea Water 耐海水		3.0%	4-5	5	4-5	5	5	5	5	Hot-Pressing 耐熱壓	Dry 乾	Immediately	3.0%	5
Post Mercerizing 後絲光		3.0%	4Y		2-3							4 hours after		
Acid Hydrolysis 酸水解		3.0%	4-5	4-5	5	5	5	5	4-5		Wet 濕	Immediately	3.0%	4-5
Perspiration 耐汗	Acid 酸	3.0%	4-5	5	5	5	5	5	5			4 hours after		
	Alkali 鹼		5	4-5	4-5	5	5	5	5	Nitrogen Oxides 氣體變褪色			3.0%	5

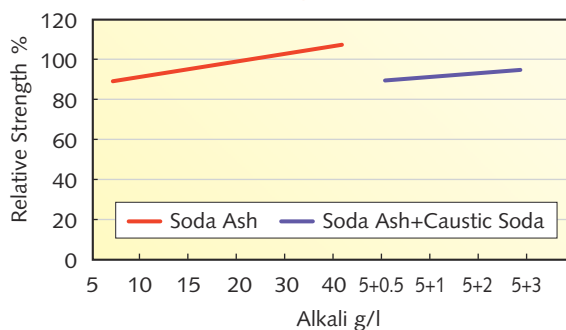
Light Fastness 日光堅牢度

Evaluation	Dyeing Depth	Color Change	ISO Blue	Evaluation		Dying Depth	Color Change	Grade ISO
Fastness	染色濃度	20 AFU	Scale	Fastness		染色濃度	20 AFU	Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.1%	2-3	4	Perspiration Light 耐汗日光	Acid	3.0%	2-3	5-6
	0.3%	3	4		Alkali		4	3-4
	1.0%	4	5					
	3.0%	4-5	5-6					
	6.0%	4-5	6					

浴比依存性 Dependency on Liquor Ratio



鹼依存性 Dependency on Alkali

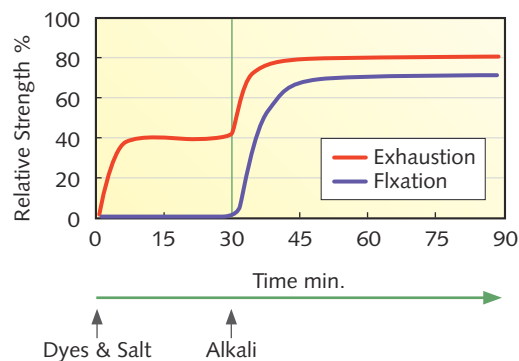


Everzol Light Red ERC

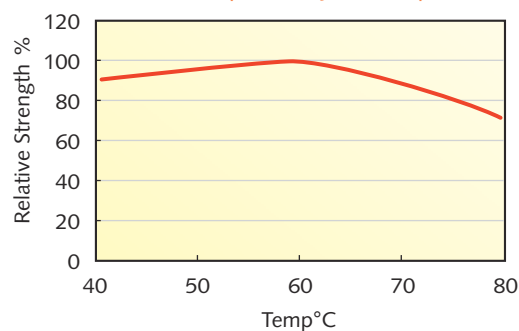
Dyeing Properties 特性

Characteristics		Evaluation
Reactivity 反應性		II
Substantivity 親和性		B
Migration 移染性		4
Washing Off 洗淨性		4
Solubility g/l 溶解度	50°C Water	100
	50°C Water + 50 g/l Salt	100
High Temp. Dyeing 高溫染色性	Dyed Fabric 色布	3-4
	Dye Liquor 染浴	4-5
Color Constancy 色恆性	Tungsten Lamp	Y
	TL 84 Lamp	Y
Metallic Ions 金屬離子	Fe ²⁺ 鐵	4-5
	Cu ²⁺ 銅	2-3
Resin Finishing 樹脂加工	Melamine 三聚氰胺	4-5
	Glyoxal 乙二醛	4-5
Shade Change after Fixing Treatment 固色後變色性	Polyamine 聚胺類	4-5
	Quaternary Ammonium Salt 四級銨鹽	4-5
Stripping 脫色性	(1)	X
	(2)	X
	(3)	X
	(4)	X
Discharge Ability 拔染性	Neutral 中性	4-5
	Alkaline 鹼性	4-5

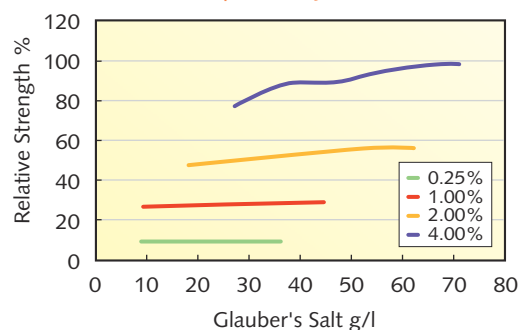
吸盡固著曲線 Exhaustion & Fixation



溫度依存性 Dependency on Temperature



芒硝依存性 Dependency on Glauber's Salt



Everzol Light Red ERC

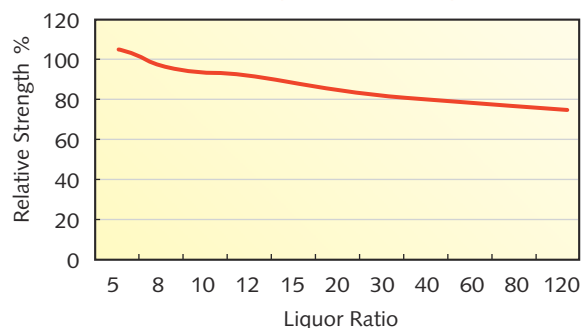
Fastness before Fixing Treatment 固色前牢度表

Evaluation		Depth of Shade 染色濃度	Color Change 變退色	Staining 污染						Evaluation			Depth of Shade 染色濃度	Grade 級數	
Fastness				Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛	Fastness					
Washing 耐水洗	ISO C10 B	3.0%	4-5	5	5	5	5	5	5	Chlorinated Water 耐氯水 (20ppm)			3.0%	3-4	
	ISO C10 C		4-5	5	5	5	5	5	5	Peroxide Bleaching 雙氧水漂白				4	
	ISO C10 D		4-5	5	4-5	5	5	5	4	Rubbing 耐摩擦	Dry 乾	3.0%	5		
	ISO C06-C1S		4-5	5	5	5	5	5	5		Wet 濕		2-3		
	ISO C06-C2S		4-5	5	5	5	5	5	5	M&S C9A 擬濕日光 (Detergent Light)			0.3%	4-5	
	M&S C4A		4-5	5	5	5	5	5	5				3.0%	4-5	
Water 耐水		3.0%	4-5	5	5	4-5	5	5	5	Multi-Washing 耐多次水洗			3.0%	4-5	
Hot Water 耐熱水		3.0%	4-5	5	4-5	5	5	5	5	Saliva (DIN) 耐唾液			3.0%	5	
Sea Water 耐海水		3.0%	4-5	5	4-5	5	5	5	5	Hot-Pressing 耐熱壓	Dry 乾	Immediately	3.0%	2-3	
Post Mercerizing 後絲光		3.0%	3-4		4-5							4 hours after		4-5	
Acid Hydrolysis 酸水解		3.0%	5	5	4-5	4-5	5	5	5			Wet 濕	Immediately	3.0%	2
Perspiration 耐汗		3.0%	4-5	5	5	5	5	5	5				4 hours after		4-5
			4-5	5	5	5	5	5	5	Nitrogen Oxides 氣體變褪色			3.0%	5	

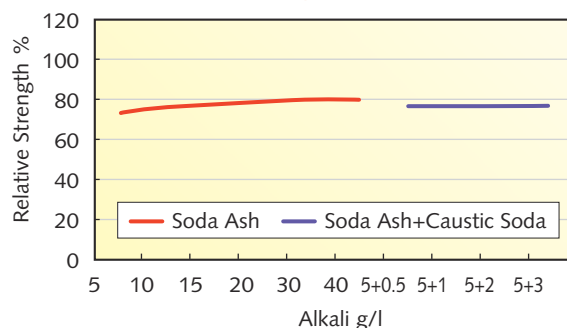
Light Fastness 日光堅牢度

Evaluation		Dyeing Depth 染色濃度	Color Change 20 AFU	ISO Blue Scale	Evaluation		Dying Depth 染色濃度	Color Change 20 AFU	Grade ISO Blue Scale
Fastness					Fastness				
Artificial Light (Xenon-Arc Lamp) 耐日光	0.1%	3	5	Perspiration Light 耐汗日光	Acid	3.0%	3-4	5-6	
	0.3%	3-4	5-6		Alkali		4	5	
	1.0%	4	6						
	3.0%	4-5	>6						
	6.0%	4-5	>6						

浴比依存性 Dependency on Liquor Ratio

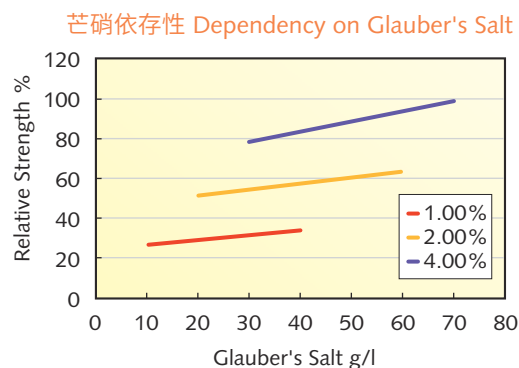
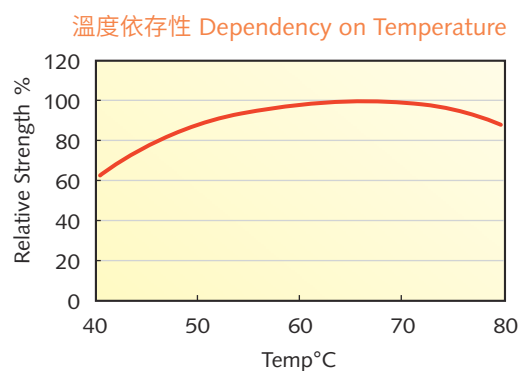
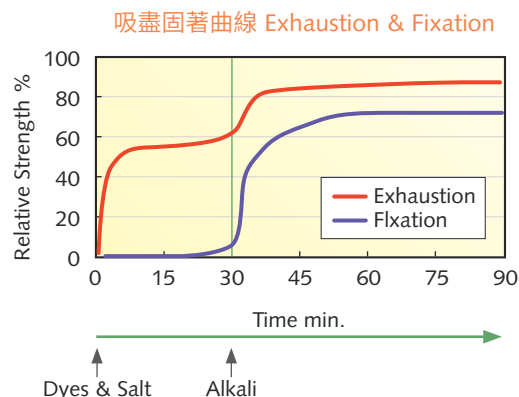


鹼依存性 Dependency on Alkali



Dyeing Properties 特性

Characteristics		Evaluation
Reactivity 反應性		II
Substantivity 親和性		B
Migration 移染性		5
Washing Off 洗淨性		4
Solubility g/l 溶解度	50°C Water	100
	50°C Water + 50 g/l Salt	100
High Temp. Dyeing 高溫染色性	Dyed Fabric 色布	1-2
	Dye Liquor 染浴	1-2
Color Constancy 色恆性	Tungsten Lamp	Y
	TL 84 Lamp	Y
Metallic Ions 金屬離子	Fe ²⁺ 鐵	3-4
	Cu ²⁺ 銅	2-3B
Resin Finishing 樹脂加工	Melamine 三聚氰胺	4-5
	Glyoxal 乙二醛	5
Shade Change after Fixing Treatment 固色後變色性	Polyamine 聚胺類	4B
	Quaternary Ammonium Salt 四級銨鹽	5
Stripping 脫色性	(1)	X
	(2)	△
	(3)	△
	(4)	△
Discharge Ability 拔染性	Neutral 中性	1-2
	Alkaline 鹼性	2



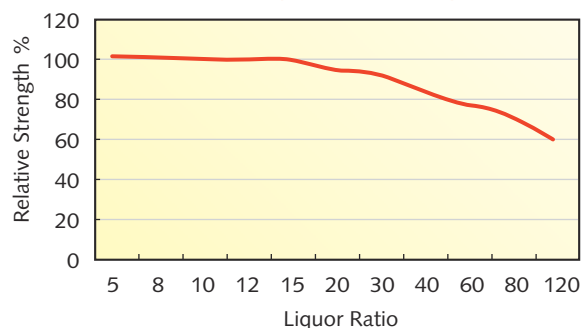
Fastness before Fixing Treatment 固色前牢度表

Evaluation		Depth of Shade 染色濃度	Color Change 變退色	Staining 污染						Evaluation			Depth of Shade 染色濃度	Grade 級數
Fastness				Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛	Fastness				
Washing 耐水洗	ISO C10 B	3.0%	5	5	5	5	5	5	5	Chlorinated Water 耐氯水 (20ppm)			3.0%	5
	ISO C10 C		5	5	5	5	5	5	5	Peroxide Bleaching 雙氧水漂白				4
	ISO C10 D		5	5	3	5	5	5	4	Rubbing 耐摩擦	Dry 乾	3.0%	4-5	
	ISO C06-C1S		5	5	4-5	5	5	5	5		Wet 濕		2-3	
	ISO C06-C2S		4-5	5	5	5	5	5	5	M&S C9A 擬濕日光 (Detergent Light)			0.3%	>4-5
	M&S C4A		4-5	5	4-5	5	5	5	5				3.0%	>4-5
Water 耐水		3.0%	5	5	4-5	4-5	5	5	5	Multi-Washing 耐多次水洗			3.0%	4-5
Hot Water 耐熱水		3.0%	5	5	5	5	5	5	5	Saliva (DIN) 耐唾液			3.0%	4-5
Sea Water 耐海水		3.0%	4-5	5	5	5	5	5	5	Hot-Pressing 耐熱壓	Dry 乾	Immediately	3.0%	4
Post Mercerizing 後絲光		3.0%	1-2Y		3-4						4 hours after	3.0%	4-5	
Acid Hydrolysis 酸水解		3.0%	4-5	5	4	4-5	5	5	4		Wet 濕	Immediately	3.0%	4
Perspiration 耐汗		3.0%	4-5	5	5	5	5	5	5		4 hours after	3.0%	4-5	
			5	5	4-5	5	5	4-5	5	Nitrogen Oxides 氣體變褪色			3.0%	5

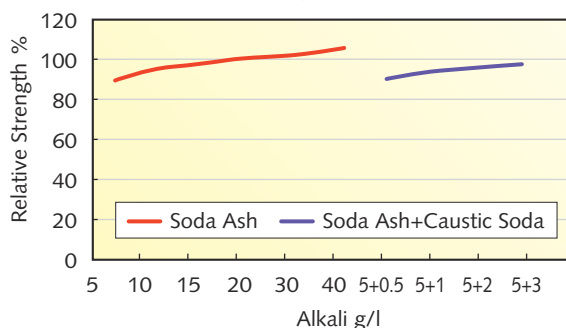
Light Fastness 日光堅牢度

Evaluation		Dyeing Depth 染色濃度	Color Change 20 AFU	ISO Blue Scale	Evaluation		Dying Depth 染色濃度	Color Change 20 AFU	Grade ISO Blue Scale
Fastness					Fastness				
Artificial Light (Xenon-Arc Lamp) 耐日光	0.1%	2-3	4	Perspiration Light 耐汗日光	Acid	3.0%	2-3	5	
	0.3%	3	4		Alkali		3	3	
	1.0%	4	5-6						
	3.0%	4-5	5-6						
	6.0%	4-5	6						

浴比依存性 Dependency on Liquor Ratio



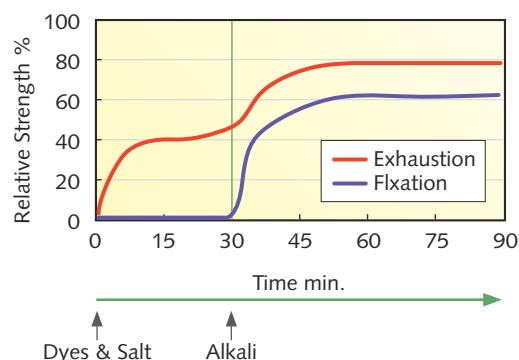
鹼依存性 Dependency on Alkali



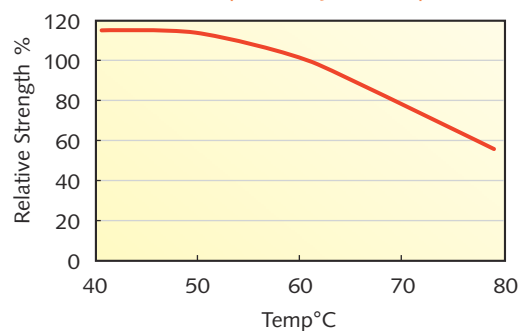
Dyeing Properties 特性

Characteristics		Evaluation
Reactivity 反應性		II
Substantivity 親和性		B
Migration 移染性		4
Washing Off 洗淨性		4
Solubility g/l 溶解度	50°C Water	100
	50°C Water + 50 g/l Salt	100
High Temp. Dyeing 高溫染色性	Dyed Fabric 色布	4
	Dye Liquor 染浴	4-5
Color Constancy 色恆性	Tungsten Lamp	R
	TL 84 Lamp	Y
Metallic Ions 金屬離子	Fe ²⁺ 鐵	4-5
	Cu ²⁺ 銅	2-3
Resin Finishing 樹脂加工	Melamine 三聚氰胺	4
	Glyoxal 乙二醛	3-4
Shade Change after Fixing Treatment 固色後變色性	Polyamine 聚胺類	4-5
	Quaternary Ammonium Salt 四級銨鹽	4-5
Stripping 脫色性	(1)	△
	(2)	△
	(3)	○
	(4)	○
Discharge Ability 拔染性	Neutral 中性	4-5
	Alkaline 鹼性	4-5

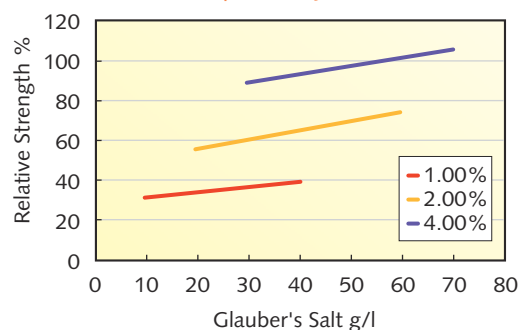
吸盡固著曲線 Exhaustion & Fixation



溫度依存性 Dependency on Temperature



芒硝依存性 Dependency on Glauber's Salt



Everzol Burnt Orange ERC

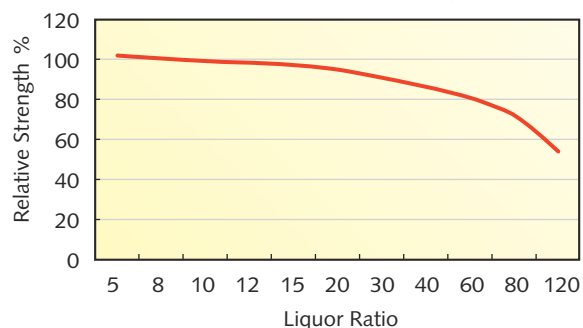
Fastness before Fixing Treatment 固色前牢度表

Evaluation		Depth of Shade 染色濃度	Color Change 變退色	Staining 污染						Evaluation			Depth of Shade 染色濃度	Grade 級數
Fastness				Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛	Fastness				
Washing 耐水洗	ISO C10 B	3.0%	4-5	5	4	5	5	5	5	Chlorinated Water 耐氯水 (20ppm)			3.0%	2-3
	ISO C10 C		4-5	5	4	5	5	5	5	Peroxide Bleaching 雙氧水漂白				3
	ISO C10 D		4	5	3-4	4-5	5	5	2-3	Rubbing 耐摩擦	Dry 乾	3.0%	4-5	
	ISO C06-C1S		4-5	5	4	5	5	5	5		Wet 濕		2-3	
	ISO C06-C2S		4-5	5	4	5	5	5	5	M&S C9A 擬濕日光 (Detergent Light)			0.3%	4-5
	M&S C4A		4-5	5	4-5	5	5	5	5				3.0%	4-5
Water 耐水		3.0%	4-5	5	5	5	5	5	5	Multi-Washing 耐多次水洗			3.0%	4
Hot Water 耐熱水		3.0%	4-5	5	5	3	5	5	5	Saliva (DIN) 耐唾液			3.0%	4-5
Sea Water 耐海水		3.0%	4-5	5	5	5	5	5	5	Hot-Pressing 耐熱壓	Dry 乾	Immediately	3.0%	4
Post Mercerizing 後絲光		3.0%	3-4R		3-4				4 hours after			5		
Acid Hydrolysis 酸水解		3.0%	4-5	5	5	5	5	5	5		Wet 濕	Immediately	3.0%	4
Perspiration 耐汗	Acid 酸	3.0%	4-5	5	5	5	5	5	5			4 hours after		5
	Alkali 鹼		4-5	5	4-5	5	5	5	5	Nitrogen Oxides 氣體變褪色			3.0%	4-5

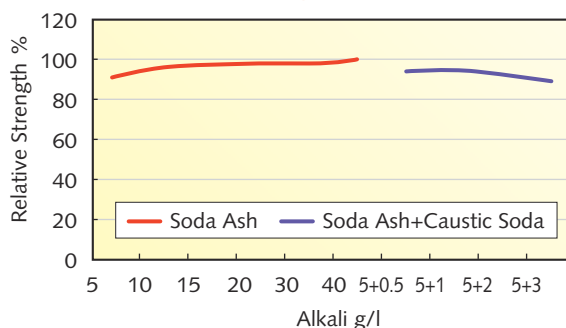
Light Fastness 日光堅牢度

Evaluation		Dyeing Depth 染色濃度	Color Change 20 AFU	ISO Blue Scale	Evaluation		Dying Depth 染色濃度	Color Change 20 AFU	Grade ISO Blue Scale
Fastness									
Artificial Light (Xenon-Arc Lamp) 耐日光	0.1%	3-4	4-5	Perspiration Light 耐汗日光	Acid	3.0%	4-5	6	
	0.3%	4	5		Alkali		4-5	6	
	1.0%	4-5	5-6						
	3.0%	4-5	6						
	6.0%	4-5	6						

浴比依存性 Dependency on Liquor Ratio



鹼依存性 Dependency on Alkali

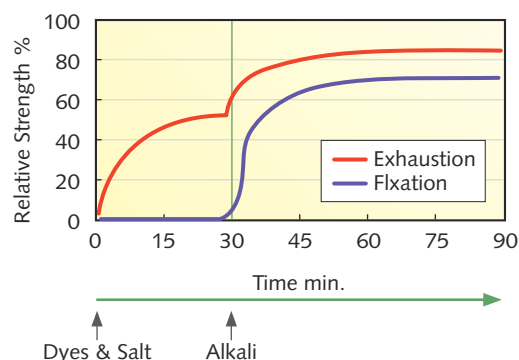


Everzol Deep Orange ERC

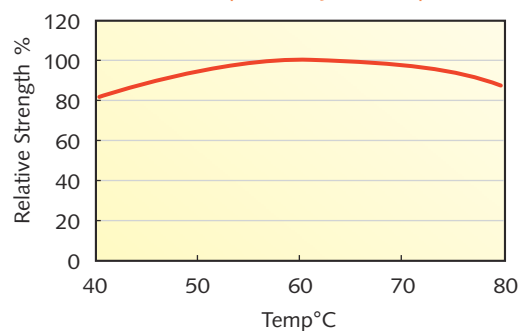
Dyeing Properties 特性

Characteristics		Evaluation
Reactivity 反應性		II
Substantivity 親和性		B
Migration 移染性		4
Washing Off 洗淨性		3
Solubility g/l 溶解度	50°C Water	100
	50°C Water + 50 g/l Salt	100
High Temp. Dyeing 高溫染色性	Dyed Fabric 色布	4
	Dye Liquor 染浴	2-3
Color Constancy 色恆性	Tungsten Lamp	Y(R)
	TL 84 Lamp	Y(G)
Metallic Ions 金屬離子	Fe ²⁺ 鐵	4-5
	Cu ²⁺ 銅	1-2
Resin Finishing 樹脂加工	Melamine 三聚氰胺	4
	Glyoxal 乙二醛	4
Shade Change after Fixing Treatment 固色後變色性	Polyamine 聚胺類	4-5
	Quaternary Ammonium Salt 四級銨鹽	4-5
Stripping 脫色性	(1)	X
	(2)	△
	(3)	△
	(4)	△
Discharge Ability 拔染性	Neutral 中性	3
	Alkaline 鹼性	3-4

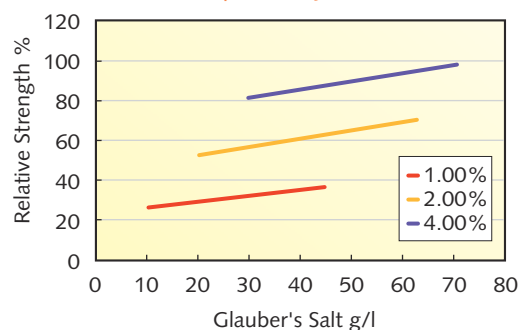
吸盡固著曲線 Exhaustion & Fixation



溫度依存性 Dependency on Temperature



芒硝依存性 Dependency on Glauber's Salt



Everzol Deep Orange ERC

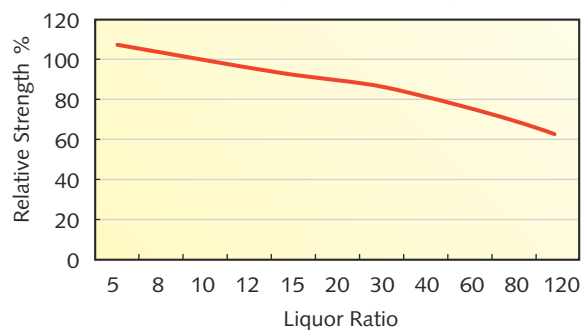
Fastness before Fixing Treatment 固色前牢度表

Evaluation		Depth of Shade 染色濃度	Color Change 變退色	Staining 污染						Evaluation		Depth of Shade 染色濃度	Grade 級數	
Fastness				Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛	Fastness				
Washing 耐水洗	ISO C10 B	3.0%	4-5	5	4-5	5	5	5	5	Chlorinated Water 耐氯水 (20ppm)		3.0%	4-5	
	ISO C10 C		4-5	5	4-5	5	5	5	5	Peroxide Bleaching 雙氧水漂白			3-4	
	ISO C10 D		4-5	5	4	5	5	5	4	Rubbing 耐摩擦	Dry 乾	3.0%	4-5	
	ISO C06-C1S		4-5	5	4-5	5	5	5	5		Wet 濕		3	
	ISO C06-C2S		4-5	5	4-5	5	5	5	5	M&S C9A 擬濕日光 (Detergent Light)		0.3%	4-5	
	M&S C4A		4-5	5	4-5	5	5	5	5			3.0%	4-5	
Water 耐水		3.0%	4-5	5	4	4	5	4	4	Multi-Washing 耐多次水洗		3.0%	4	
Hot Water 耐熱水		3.0%	4-5	5	4-5	4-5	5	5	5	Saliva (DIN) 耐唾液		3.0%	4-5	
Sea Water 耐海水		3.0%	4-5	5	4-5	5	5	5	5	Hot-Pressing 耐熱壓	Dry 乾	Immediately	3.0%	3-4
Post Mercerizing 後絲光		3.0%	4		3-4						4 hours after		4-5	
Acid Hydrolysis 酸水解		3.0%	4-5	5	3-4	3-4	5	5	4-5		Wet 濕	Immediately	3.0%	3-4
Perspiration 耐汗	Acid 酸	3.0%	4-5	5	4-5	4-5	5	5	5		4 hours after			4-5
	Alkali 鹼		4-5	5	4-5	4-5	5	5	5	Nitrogen Oxides 氣體變褪色		3.0%	5	

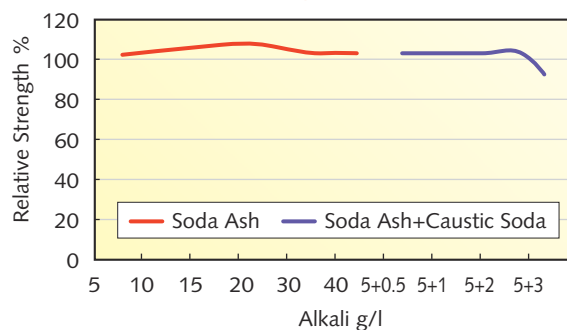
Light Fastness 日光堅牢度

Evaluation	Dyeing Depth	Color Change	ISO Blue	Evaluation		Dying Depth	Color Change	Grade ISO
Fastness	染色濃度	20 AFU	Scale	Fastness		染色濃度	20 AFU	Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.1%	3	4-5	Perspiration Light 耐汗日光	Acid	3.0%	4	5-6
	0.3%	3-4	4-5		Alkali		4	5-6
	1.0%	4	5					
	3.0%	4-5	5-6					
	6.0%	4-5	6					

浴比依存性 Dependency on Liquor Ratio



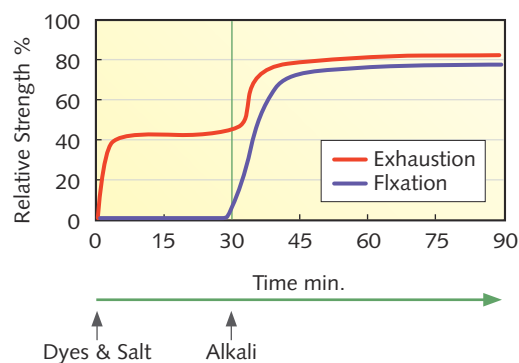
鹼依存性 Dependency on Alkali



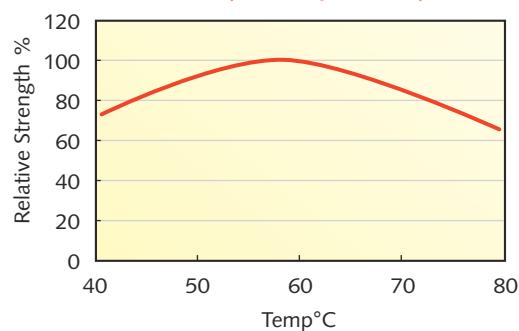
Dyeing Properties 特性

Characteristics		Evaluation
Reactivity 反應性		II
Substantivity 親和性		B
Migration 移染性		4
Washing Off 洗淨性		4
Solubility g/l 溶解度	50°C Water	100
	50°C Water + 50 g/l Salt	100
High Temp. Dyeing 高溫染色性	Dyed Fabric 色布	3-4
	Dye Liquor 染浴	3-4R
Color Constancy 色恆性	Tungsten Lamp	(Y)
	TL 84 Lamp	G(Y)
Metallic Ions 金屬離子	Fe ²⁺ 鐵	4-5
	Cu ²⁺ 銅	3
Resin Finishing 樹脂加工	Melamine 三聚氰胺	2-3
	Glyoxal 乙二醛	4-5
Shade Change after Fixing Treatment 固色後變色性	Polyamine 聚胺類	4-5
	Quaternary Ammonium Salt 四級銨鹽	4-5
Stripping 脫色性	(1)	X
	(2)	X
	(3)	X
	(4)	X
Discharge Ability 拔染性	Neutral 中性	3
	Alkaline 鹼性	3

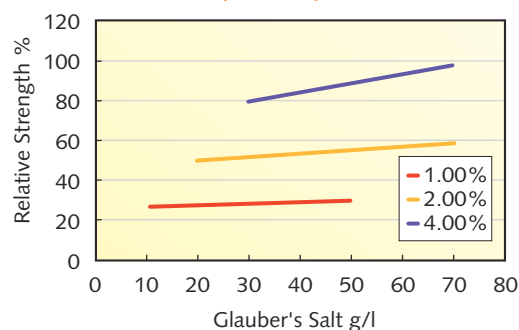
吸盡固著曲線 Exhaustion & Fixation



溫度依存性 Dependency on Temperature



芒硝依存性 Dependency on Glauber's Salt



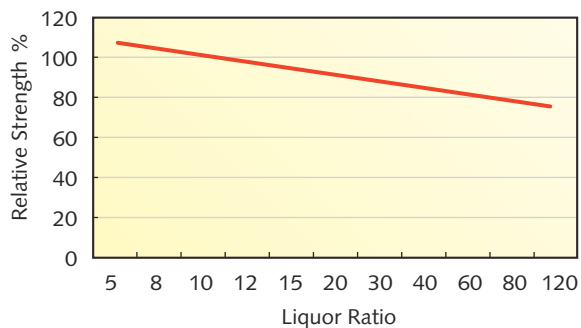
Fastness before Fixing Treatment 固色前牢度表

Evaluation		Depth of Shade 染色濃度	Color Change 變退色	Staining 污染						Evaluation			Depth of Shade 染色濃度	Grade 級數
				Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛					
Fastness										Fastness				
Washing 耐水洗	ISO C10 B	3.0%	4-5	5	5	5	5	5	5	Chlorinated Water 耐氯水 (20ppm)		3.0%	4	
	ISO C10 C		4-5	5	5	5	5	5	5	Peroxide Bleaching 雙氧水漂白			4-5	
	ISO C10 D		4-5	5	4-5	5	5	5	5	Rubbing 耐摩擦	Dry 乾	3.0%	4-5	
	ISO C06-C1S		4-5	5	4-5	5	5	5	5		Wet 濕		2	
	ISO C06-C2S		4-5	5	4-5	5	5	5	5	M&S C9A 擬濕日光 (Detergent Light)		0.3%	4-5	
	M&S C4A		4-5	5	5	5	5	5	5			3.0%	5	
Water 耐水		3.0%	4-5	5	5	5	5	5	5	Multi-Washing 耐多次水洗		3.0%	4	
Hot Water 耐熱水		3.0%	4-5	5	5	5	5	5	5	Saliva (DIN) 耐唾液		3.0%	5	
Sea Water 耐海水		3.0%	4-5	5	5	5	5	5	5	Hot- Pressing 耐熱壓	Dry 乾	Immediately	3.0%	3-4
Post Mercerizing 後絲光		3.0%	3-4		2							4 hours after		5
Acid Hydrolysis 酸水解		3.0%	4-5	5	4-5	4	5	5	4-5		Wet 濕	Immediately	3.0%	3
Perspiration 耐汗	Acid 酸	3.0%	4-5	5	5	5	5	5	5			4 hours after		4-5
	Alkali 鹼		4-5	5	5	5	5	5	5	Nitrogen Oxides 氣體變褪色		3.0%	4-5	

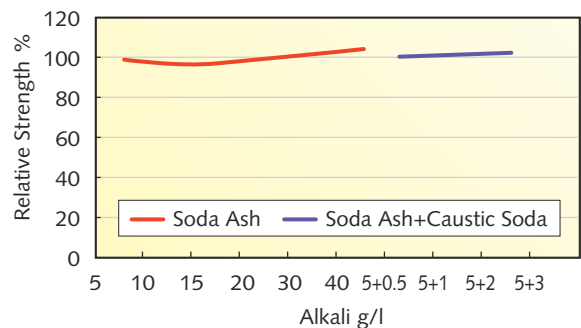
Light Fastness 日光堅牢度

Evaluation	Dyeing Depth	Color Change	ISO Blue	Evaluation		Dyeing Depth	Color Change	Grade ISO
Fastness	染色濃度	20 AFU	Scale	Fastness		染色濃度	20 AFU	Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.1%	4	5-6	Perspiration Light 耐汗日光	Acid	3.0%	3-4	6
	0.3%	4-5	6		Alkali		4	5
	1.0%	4-5	>6					
	3.0%	4-5	>6					
	6.0%	4-5	>6					

浴比依存性 Dependency on Liquor Ratio

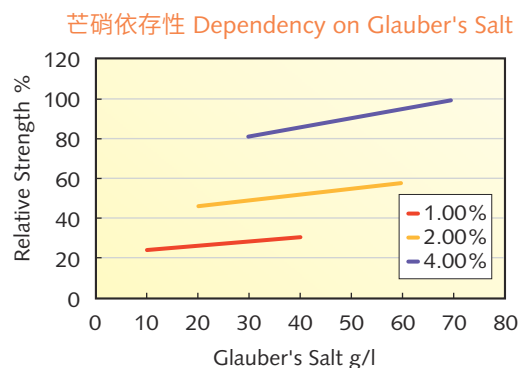
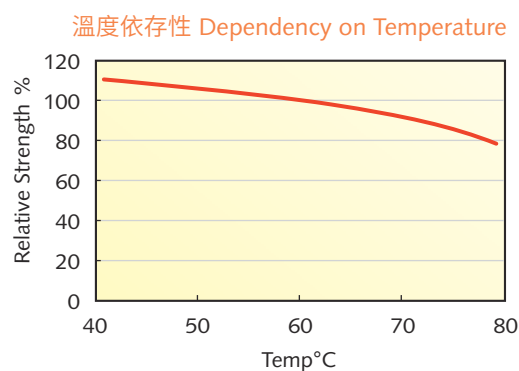
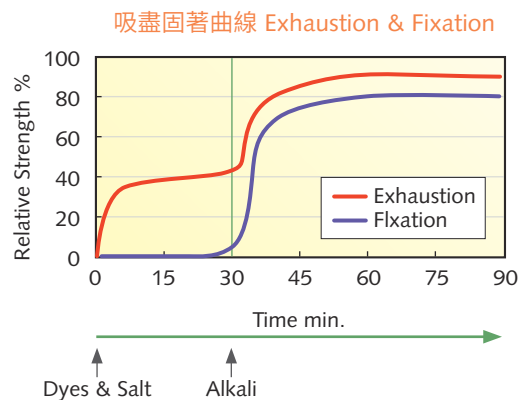


鹼依存性 Dependency on Alkali



Dyeing Properties 特性

Characteristics		Evaluation
Reactivity 反應性		II
Substantivity 親和性		B
Migration 移染性		4
Washing Off 洗淨性		4-5
Solubility g/l 溶解度	50°C Water	100
	50°C Water + 50 g/l Salt	100
High Temp. Dyeing 高溫染色性	Dyed Fabric 色布	4-5
	Dye Liquor 染浴	4-5
Color Constancy 色恆性	Tungsten Lamp	G(B)
	TL 84 Lamp	Y(R)
Metallic Ions 金屬離子	Fe ²⁺ 鐵	5
	Cu ²⁺ 銅	3
Resin Finishing 樹脂加工	Melamine 三聚氰胺	4-5
	Glyoxal 乙二醛	4-5
Shade Change after Fixing Treatment 固色後變色性	Polyamine 聚胺類	4-5
	Quaternary Ammonium Salt 四級銨鹽	4-5
Stripping 脫色性	(1)	X
	(2)	△
	(3)	○
	(4)	△
Discharge Ability 拔染性	Neutral 中性	3
	Alkaline 鹼性	4-5



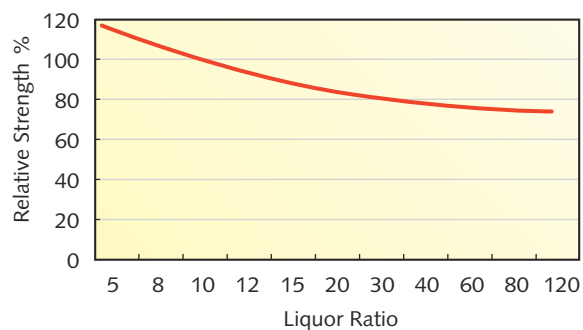
Fastness before Fixing Treatment 固色前牢度表

Evaluation		Depth of Shade 染色濃度	Color Change 變退色	Staining 污染						Evaluation			Depth of Shade 染色濃度	Grade 級數
Fastness				Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛	Fastness				
Washing 耐水洗	ISO C10 B	3.0%	4-5	5	4-5	5	5	5	5	Chlorinated Water 耐氯水 (20ppm)			3.0%	4
	ISO C10 C		4-5	5	4-5	5	5	5	5	Peroxide Bleaching 雙氧水漂白				5
	ISO C10 D		4-5	5	3-4	5	5	5	4	Rubbing 耐摩擦	Dry 乾	3.0%	4-5	
	ISO C06-C1S		4-5	5	4-5	5	5	5	5		Wet 濕		2-3	
	ISO C06-C2S		4-5	5	4-5	5	5	5	5	M&S C9A 擬濕日光 (Detergent Light)			0.3%	4-5
	M&S C4A		4-5	5	5	5	5	5	5				3.0%	4-5
Water 耐水		3.0%	4-5	5	4-5	4-5	4-5	4-5	5	Multi-Washing 耐多次水洗			3.0%	4-5
Hot Water 耐熱水		3.0%	4-5	5	5	5	5	5	5	Saliva (DIN) 耐唾液			3.0%	5
Sea Water 耐海水		3.0%	4-5	5	4-5	5	5	4-5	5	Hot-Pressing 耐熱壓	Dry 乾	Immediately	3.0%	2
Post Mercerizing 後絲光		3.0%	2-3		4						4 hours after	4-5		
Acid Hydrolysis 酸水解		3.0%	4-5	5	4-5	4-5	5	5	5		Wet 濕	Immediately	3.0%	2
Perspiration 耐汗	Acid 酸	3.0%	4-5	5	4	4-5	4-5	4-5	5			4 hours after		4
	Alkali 鹼		4-5	5	4	4-5	4-5	4-5	5	Nitrogen Oxides 氣體變褪色			3.0%	5

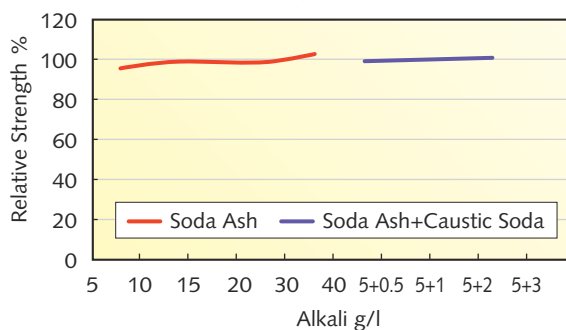
Light Fastness 日光堅牢度

Evaluation		Dyeing Depth 染色濃度	Color Change 20 AFU	ISO Blue Scale	Evaluation		Dying Depth 染色濃度	Color Change 20 AFU	Grade ISO Blue Scale
Fastness					Fastness				
Artificial Light (Xenon-Arc Lamp) 耐日光	0.1%	3	4	Perspiration Light 耐汗日光	Acid	3.0%	3-4	5-6	
	0.3%	3-4	4-5		Alkali		4	4-5	
	1.0%	4	5						
	3.0%	4-5	5-6						
	6.0%	4-5	6						

浴比依存性 Dependency on Liquor Ratio



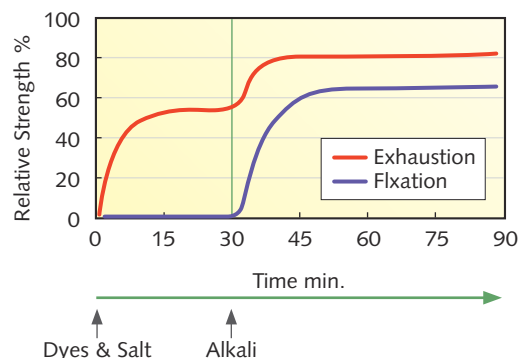
鹼依存性 Dependency on Alkali



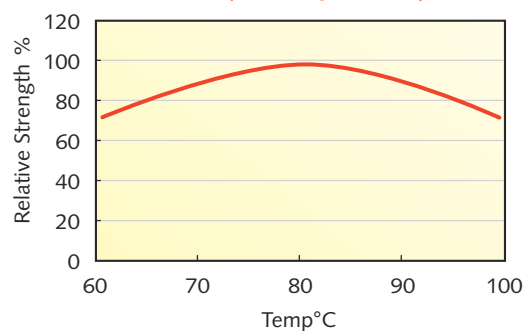
Dyeing Properties 特性

Characteristics		Evaluation
Reactivity 反應性		II
Substantivity 親和性		B
Migration 移染性		4
Washing Off 洗淨性		3
Solubility g/l 溶解度	50°C Water	100
	50°C Water + 50 g/l Salt	100
High Temp. Dyeing 高溫染色性	Dyed Fabric 色布	4D
	Dye Liquor 染浴	4Y
Color Constancy 色恆性	Tungsten Lamp	R
	TL 84 Lamp	R
Metallic Ions 金屬離子	Fe ²⁺ 鐵	5
	Cu ²⁺ 銅	5
Resin Finishing 樹脂加工	Melamine 三聚氰胺	5
	Glyoxal 乙二醛	4Y
Shade Change after Fixing Treatment 固色後變色性	Polyamine 聚胺類	3-4Y
	Quaternary Ammonium Salt 四級銨鹽	4-5
Stripping 脫色性	(1)	△
	(2)	△
	(3)	○
	(4)	○
Discharge Ability 拔染性	Neutral 中性	3
	Alkaline 鹼性	2

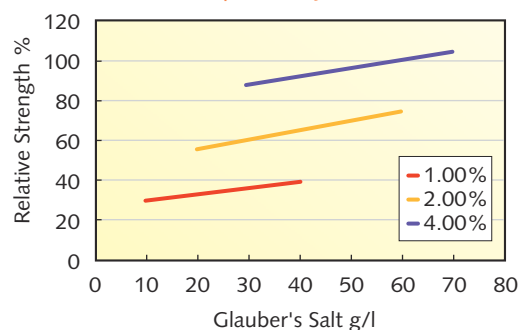
吸盡固著曲線 Exhaustion & Fixation



溫度依存性 Dependency on Temperature



芒硝依存性 Dependency on Glauber's Salt



Everzol Turquoise Blue ERC

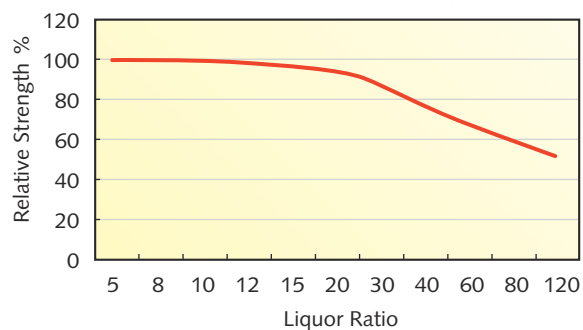
Fastness before Fixing Treatment 固色前牢度表

Evaluation		Depth of Shade 染色濃度	Color Change 變退色	Staining 污染						Evaluation			Depth of Shade 染色濃度	Grade 級數
Fastness				Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛	Fastness				
Washing 耐水洗	ISO C10 B	3.0%	5	5	3-4	5	5	5	4-5	Chlorinated Water 耐氯水 (20ppm)			3.0%	3
	ISO C10 C		4-5	5	3	4-5	5	5	4-5	Peroxide Bleaching 雙氧水漂白				2
	ISO C10 D		4-5	5	2	4-5	5	4-5	2-3	Rubbing 耐摩擦	Dry 乾	3.0%	5	
	ISO C06-C1S		4-5	5	3	5	5	4-5	5		Wet 濕		2-3	
	ISO C06-C2S		5	5	3-4	5	5	5	5	M&S C9A 擬濕日光 (Detergent Light)			0.3%	4-5
	M&S C4A		4-5	5	3	4	5	5	4-5				3.0%	4-5
Water 耐水		3.0%	5	5	5	5	5	5	5	Multi-Washing 耐多次水洗			3.0%	4
Hot Water 耐熱水		3.0%	5	5	5	5	5	5	5	Saliva (DIN) 耐唾液			3.0%	4-5
Sea Water 耐海水		3.0%	5	5	5	5	5	5	5	Hot-Pressing 耐熱壓	Dry 乾	Immediately	3.0%	4-5
Post Mercerizing 後絲光		3.0%	3G		4						4 hours after	5		
Acid Hydrolysis 酸水解		3.0%	3Y	5	5	5	5	5	5		Wet 濕	Immediately	3.0%	4-5
Perspiration 耐汗	Acid 酸	3.0%	4-5	5	5	5	5	5	5			4 hours after		4-5
	Alkali 鹼		4-5	5	5	5	5	5	5	5	Nitrogen Oxides 氣體變褪色			3.0%

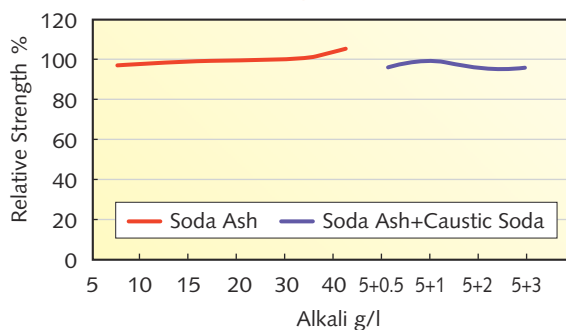
Light Fastness 日光堅牢度

Evaluation		Dyeing Depth 染色濃度	Color Change 20 AFU	ISO Blue Scale	Evaluation		Dying Depth 染色濃度	Color Change 20 AFU	Grade ISO Blue Scale
Fastness					Fastness				
Artificial Light (Xenon-Arc Lamp) 耐日光	0.1%	3-4	5	Perspiration Light 耐汗日光	Acid	3.0%	2	4	
	0.3%	4	5-6		Alkali		2-3	3	
	1.0%	4	5-6						
	3.0%	4-5	6						
	6.0%	4-5	6						

浴比依存性 Dependency on Liquor Ratio



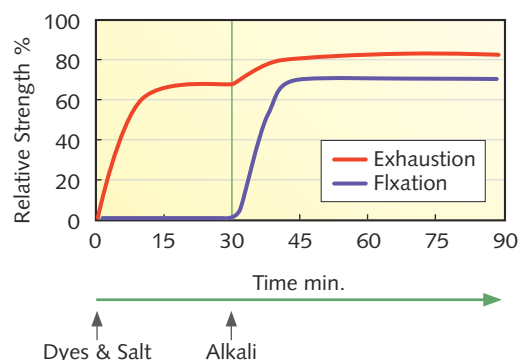
鹼依存性 Dependency on Alkali



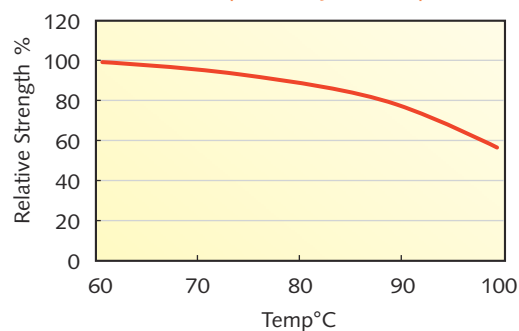
Dyeing Properties 特性

Characteristics		Evaluation
Reactivity 反應性		II
Substantivity 親和性		B
Migration 移染性		2-3
Washing Off 洗淨性		3-4
Solubility g/l 溶解度	50°C Water	100
	50°C Water + 50 g/l Salt	100
High Temp. Dyeing 高溫染色性	Dyed Fabric 色布	3
	Dye Liquor 染浴	3-4
Color Constancy 色恆性	Tungsten Lamp	R
	TL 84 Lamp	R
Metallic Ions 金屬離子	Fe ²⁺ 鐵	4-5
	Cu ²⁺ 銅	5
Resin Finishing 樹脂加工	Melamine 三聚氰胺	4-5
	Glyoxal 乙二醛	4-5
Shade Change after Fixing Treatment 固色後變色性	Polyamine 聚胺類	3-4
	Quaternary Ammonium Salt 四級銨鹽	4-5
Stripping 脫色性	(1)	X
	(2)	X
	(3)	△
	(4)	△
Discharge Ability 拔染性	Neutral 中性	1
	Alkaline 鹼性	1

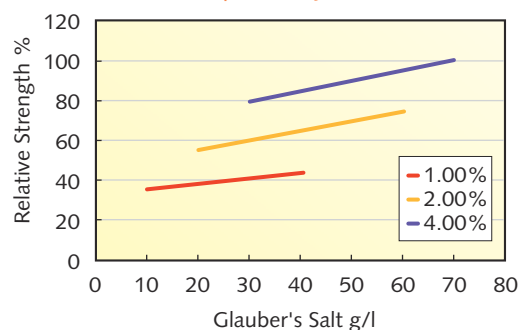
吸盡固著曲線 Exhaustion & Fixation



溫度依存性 Dependency on Temperature



芒硝依存性 Dependency on Glauber's Salt



Everzol Sky Blue ERC

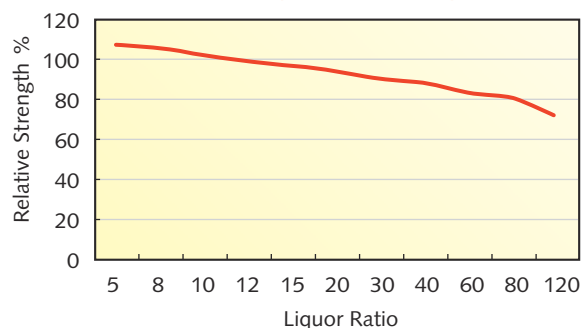
Fastness before Fixing Treatment 固色前牢度表

Evaluation		Depth of Shade 染色濃度	Color Change 變退色	Staining 污染						Evaluation		Depth of Shade 染色濃度	Grade 級數	
Fastness				Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛	Fastness				
Washing 耐水洗	ISO C10 B	3.0%	5	5	4-5	5	5	5	5	Chlorinated Water 耐氯水 (20ppm)		3.0%	1	
	ISO C10 C		5	5	4-5	5	5	5	5	Peroxide Bleaching 雙氧水漂白			3-4	
	ISO C10 D		5	5	4	5	5	5	5	Rubbing 耐摩擦	Dry 乾	3.0%	5	
	ISO C06-C1S		5	5	4	5	5	5	5		Wet 濕		2-3	
	ISO C06-C2S		5	5	4	5	5	5	5	M&S C9A 擬濕日光 (Detergent Light)		0.3%	4-5	
	M&S C4A		5	5	5	5	5	5	5			3.0%	4-5	
Water 耐水		3.0%	5	5	4-5	4-5	5	5	5	Multi-Washing 耐多次水洗		3.0%	5	
Hot Water 耐熱水		3.0%	5	5	5	5	5	5	5	Saliva (DIN) 耐唾液		3.0%	4-5	
Sea Water 耐海水		3.0%	5	5	4-5	5	5	5	5	Hot-Pressing 耐熱壓	Dry 乾	Immediately	3.0%	3-4R
Post Mercerizing 後絲光		3.0%	3G(B)		4-5						4 hours after	5		
Acid Hydrolysis 酸水解		3.0%	4-5	5	3	1-2	5	5	4		Wet 濕	Immediately	3.0%	3-4R
Perspiration 耐汗	Acid 酸	3.0%	4-5	5	4-5	4-5	5	5	5			4 hours after		5
	Alkali 鹼		5	5	4-5	5	5	5	5	Nitrogen Oxides 氣體變褪色		3.0%	2	

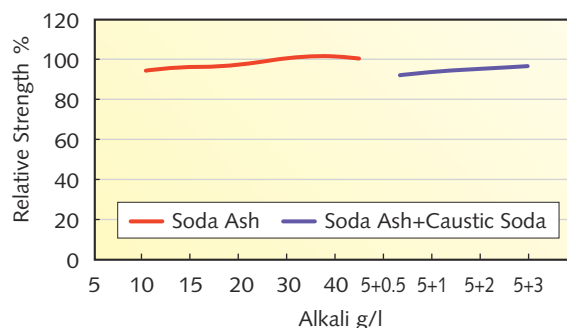
Light Fastness 日光堅牢度

Evaluation	Dyeing Depth	Color Change	ISO Blue	Evaluation		Dying Depth	Color Change	Grade ISO
Fastness	染色濃度	20 AFU	Scale	Fastness		染色濃度	20 AFU	Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.1%	4-5	6	Perspiration Light 耐汗日光	Acid	3.0%	4-5	>6
	0.3%	4-5	>6		Alkali		4-5	>6
	1.0%	4-5	>6					
	3.0%	4-5	>6					
	6.0%	4-5	>6					

浴比依存性 Dependency on Liquor Ratio



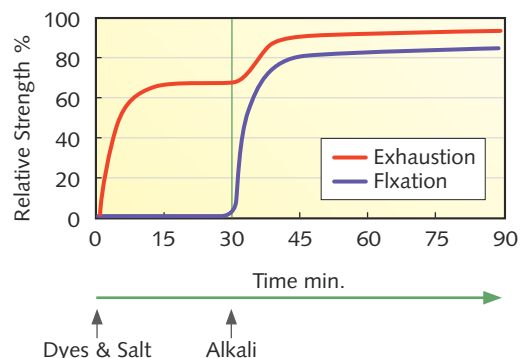
鹼依存性 Dependency on Alkali



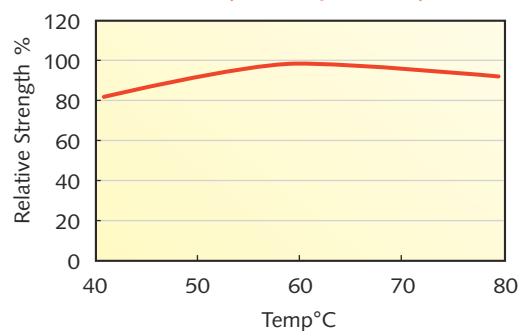
Dyeing Properties 特性

Characteristics		Evaluation
Reactivity 反應性		II
Substantivity 親和性		B
Migration 移染性		3-4
Washing Off 洗淨性		4
Solubility g/l 溶解度	50°C Water	100
	50°C Water + 50 g/l Salt	100
High Temp. Dyeing 高溫染色性	Dyed Fabric 色布	3-4
	Dye Liquor 染浴	4
Color Constancy 色恆性	Tungsten Lamp	G(B)
	TL 84 Lamp	B(G)
Metallic Ions 金屬離子	Fe ²⁺ 鐵	5
	Cu ²⁺ 銅	5
Resin Finishing 樹脂加工	Melamine 三聚氰胺	4
	Glyoxal 乙二醛	4
Shade Change after Fixing Treatment 固色後變色性	Polyamine 聚胺類	4-5
	Quaternary Ammonium Salt 四級銨鹽	4-5
Stripping 脫色性	(1)	X
	(2)	X
	(3)	X
	(4)	X
Discharge Ability 拔染性	Neutral 中性	3
	Alkaline 鹼性	3-4

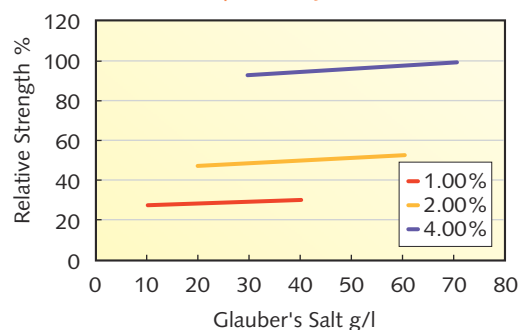
吸盡固著曲線 Exhaustion & Fixation



溫度依存性 Dependency on Temperature



芒硝依存性 Dependency on Glauber's Salt



Everzol Blue ERC

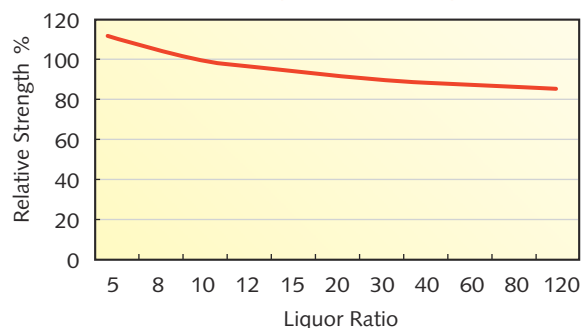
Fastness before Fixing Treatment 固色前牢度表

Evaluation		Depth of Shade 染色濃度	Color Change 變退色	Staining 污染						Evaluation			Depth of Shade 染色濃度	Grade 級數
Fastness				Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛	Fastness				
Washing 耐水洗	ISO C10 B	3.0%	4-5	5	4-5	5	5	5	5	Chlorinated Water 耐氯水 (20ppm)			3.0%	3
	ISO C10 C		4-5	5	4-5	5	5	5	5	Peroxide Bleaching 雙氧水漂白				2-3
	ISO C10 D		4-5	5	3-4	5	5	5	5	Rubbing 耐摩擦	Dry 乾	3.0%	4-5	
	ISO C06-C1S		4-5	5	4	5	5	5	5		Wet 濕		3	
	ISO C06-C2S		4-5	5	4	5	5	5	5	M&S C9A 擬濕日光 (Detergent Light)			0.3%	5
	M&S C4A		4-5	5	4-5	5	5	5	5				3.0%	5
Water 耐水		3.0%	4-5	5	4-5	4-5	5	5	5	Multi-Washing 耐多次水洗			3.0%	4-5
Hot Water 耐熱水		3.0%	4-5	5	5	5	5	5	5	Saliva (DIN) 耐唾液			3.0%	4-5
Sea Water 耐海水		3.0%	4-5	5	4-5	5	5	5	5	Hot-Pressing 耐熱壓	Dry 乾	Immediately	3.0%	4
Post Mercerizing 後絲光		3.0%	4-5		4-5						4 hours after	4-5		
Acid Hydrolysis 酸水解		3.0%	4-5	5	4-5	4-5	5	5	4-5		Wet 濕	Immediately	3.0%	4
Perspiration 耐汗	Acid 酸	3.0%	4-5	5	4-5	4	5	5	5			4 hours after		4-5
	Alkali 鹼		4-5	5	4-5	4-5	5	5	5	Nitrogen Oxides 氣體變褪色			3.0%	4-5

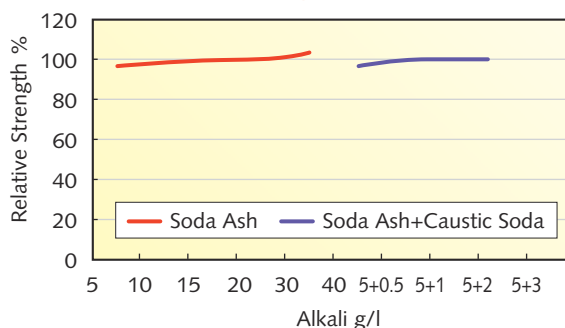
Light Fastness 日光堅牢度

Evaluation	Dyeing Depth	Color Change	ISO Blue	Evaluation		Dying Depth	Color Change	Grade ISO
Fastness	染色濃度	20 AFU	Scale	Fastness		染色濃度	20 AFU	Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.1%	4-5	6	Perspiration Light 耐汗日光	Acid	3.0%	3-4	5-6
	0.3%	4-5	6		Alkali		4	5
	1.0%	4-5	6					
	3.0%	4-5	>6					
	6.0%	4-5	>6					

浴比依存性 Dependency on Liquor Ratio



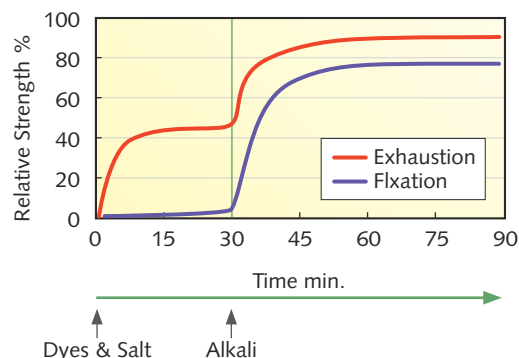
鹼依存性 Dependency on Alkali



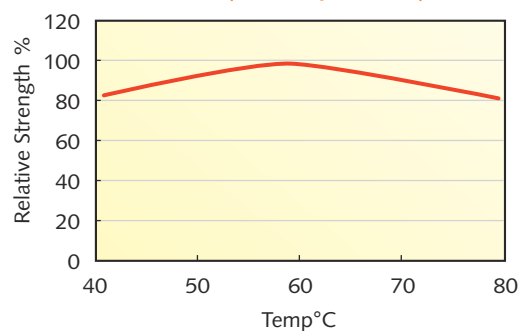
Dyeing Properties 特性

Characteristics		Evaluation
Reactivity 反應性		II
Substantivity 親和性		B
Migration 移染性		4
Washing Off 洗淨性		3-4
Solubility g/l 溶解度	50°C Water	100
	50°C Water + 50 g/l Salt	100
High Temp. Dyeing 高溫染色性	Dyed Fabric 色布	4
	Dye Liquor 染浴	2-3
Color Constancy 色恆性	Tungsten Lamp	G(B)
	TL 84 Lamp	B
Metallic Ions 金屬離子	Fe ²⁺ 鐵	5
	Cu ²⁺ 銅	3
Resin Finishing 樹脂加工	Melamine 三聚氰胺	4
	Glyoxal 乙二醛	4
Shade Change after Fixing Treatment 固色後變色性	Polyamine 聚胺類	4-5
	Quaternary Ammonium Salt 四級銨鹽	5
Stripping 脫色性	(1)	X
	(2)	X
	(3)	X
	(4)	X
Discharge Ability 拔染性	Neutral 中性	3
	Alkaline 鹼性	3-4

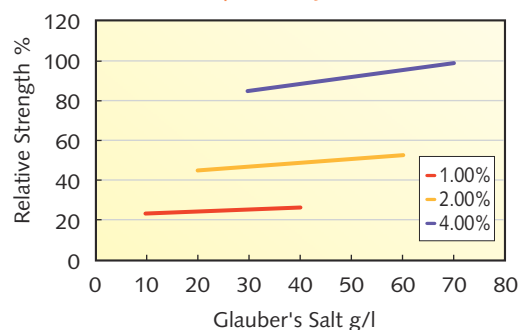
吸盡固著曲線 Exhaustion & Fixation



溫度依存性 Dependency on Temperature



芒硝依存性 Dependency on Glauber's Salt



Everzol Dark Blue ERC

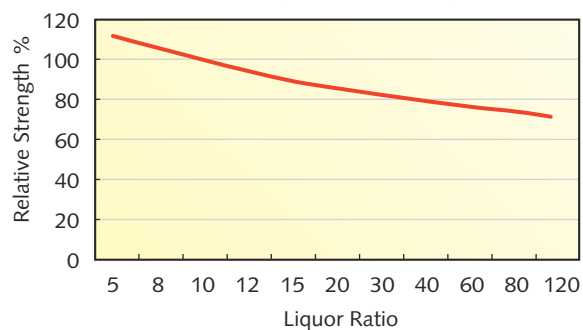
Fastness before Fixing Treatment 固色前牢度表

Evaluation		Depth of Shade 染色濃度	Color Change 變退色	Staining 污染						Evaluation			Depth of Shade 染色濃度	Grade 級數
Fastness				Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛	Fastness				
Washing 耐水洗	ISO C10 B	3.0%	4-5	5	5	5	5	5	5	Chlorinated Water 耐氯水 (20ppm)			3.0%	3
	ISO C10 C		4-5	5	4-5	5	5	5	5	Peroxide Bleaching 雙氧水漂白				2-3
	ISO C10 D		4	5	3	5	5	5	3	Rubbing 耐摩擦	Dry 乾	3.0%	5	
	ISO C06-C1S		4-5	5	4-5	5	5	5	5		Wet 濕		3	
	ISO C06-C2S		4-5	5	4-5	5	5	5	5	M&S C9A 擬濕日光 (Detergent Light)			0.3%	4
	M&S C4A		4-5	5	5	5	5	5	5				3.0%	4-5
Water 耐水		3.0%	5	5	5	5	5	5	5	Multi-Washing 耐多次水洗			3.0%	4
Hot Water 耐熱水		3.0%	4-5	5	5	5	5	5	5	Saliva (DIN) 耐唾液			3.0%	5
Sea Water 耐海水		3.0%	4-5	5	5	5	5	5	5	Hot-Pressing 耐熱壓	Dry 乾	Immediately	3.0%	4-5
Post Mercerizing 後絲光		3.0%	3-4		4							4 hours after		5
Acid Hydrolysis 酸水解		3.0%	4-5	5	4-5	5	5	5	5		Wet 濕	Immediately	3.0%	4-5
Perspiration 耐汗	Acid 酸	3.0%	4-5	5	4-5	5	5	5	5			4 hours after		5
	Alkali 鹼		4-5	5	4-5	5	5	5	5	Nitrogen Oxides 氣體變褪色			3.0%	4-5

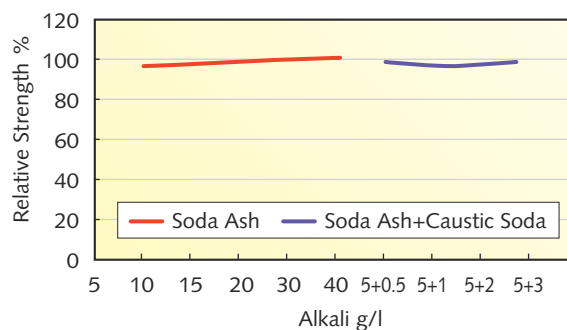
Light Fastness 日光堅牢度

Evaluation	Dyeing Depth	Color Change	ISO Blue	Evaluation		Dying Depth	Color Change	Grade ISO
Fastness	染色濃度	20 AFU	Scale	Fastness		染色濃度	20 AFU	Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.1%	2	3-4	Perspiration Light 耐汗日光	Acid	3.0%	2-3	4-5
	0.3%	3	4		Alkali		3	4
	1.0%	3-4	4-5					
	3.0%	4-5	5					
	6.0%	5	6					

浴比依存性 Dependency on Liquor Ratio



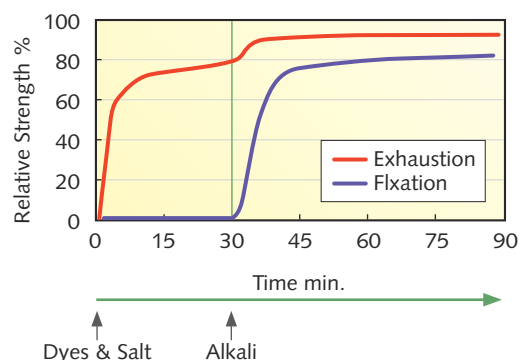
鹼依存性 Dependency on Alkali



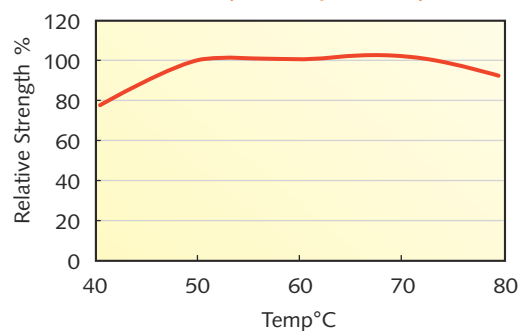
Dyeing Properties 特性

Characteristics		Evaluation
Reactivity 反應性		II
Substantivity 親和性		A
Migration 移染性		4-5
Washing Off 洗淨性		3-4
Solubility g/l 溶解度	50°C Water	100
	50°C Water + 50 g/l Salt	100
High Temp. Dyeing 高溫染色性	Dyed Fabric 色布	2-3
	Dye Liquor 染浴	2-3
Color Constancy 色恆性	Tungsten Lamp	1-2R
	TL 84 Lamp	2R
Metallic Ions 金屬離子	Fe ²⁺ 鐵	4-5
	Cu ²⁺ 銅	3G
Resin Finishing 樹脂加工	Melamine 三聚氰胺	4-5
	Glyoxal 乙二醛	5
Shade Change after Fixing Treatment 固色後變色性	Polyamine 聚胺類	4-5
	Quaternary Ammonium Salt 四級銨鹽	5
Stripping 脫色性	(1)	X
	(2)	X
	(3)	△
	(4)	X
Discharge Ability 拔染性	Neutral 中性	4-5
	Alkaline 鹼性	3-4

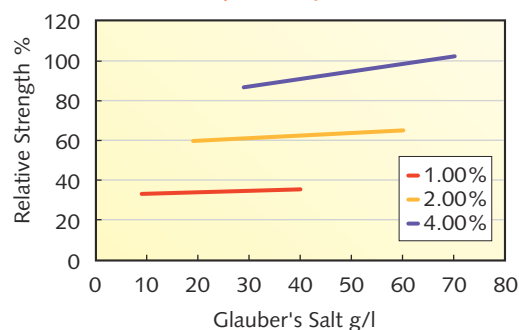
吸盡固著曲線 Exhaustion & Fixation



溫度依存性 Dependency on Temperature



芒硝依存性 Dependency on Glauber's Salt



Everzol Navy ERC

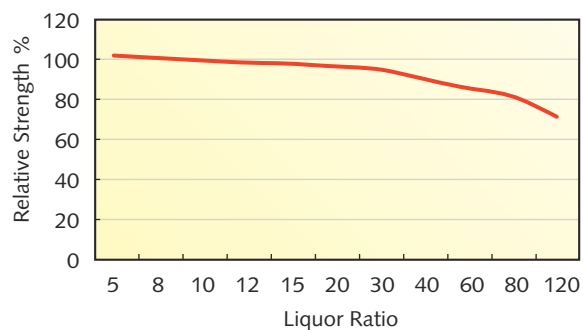
Fastness before Fixing Treatment 固色前牢度表

Evaluation		Depth of Shade 染色濃度	Color Change 變退色	Staining 污染						Evaluation			Depth of Shade 染色濃度	Grade 級數
Fastness				Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛	Fastness				
Washing 耐水洗	ISO C10 B	3.0%	5	5	4-5	5	5	5	5	Chlorinated Water 耐氯水 (20ppm)			3.0%	4-5
	ISO C10 C		4-5	5	4-5	5	5	5	5	Peroxide Bleaching 雙氧水漂白				
	ISO C10 D		5	5	4	5	5	5	4	Rubbing 耐摩擦	Dry 乾	3.0%	5	
	ISO C06-C1S		4-5	5	4-5	5	5	5	5		Wet 濕			
	ISO C06-C2S		4-5	5	5	5	5	5	5	M&S C9A 擬濕日光 (Detergent Light)			0.3%	>4-5
	M&S C4A		4-5	5	5	5	5	5	5				3.0%	>4-5
Water 耐水		3.0%	5	5	4-5	4-5	5	5	4	Multi-Washing 耐多次水洗			3.0%	4-5
Hot Water 耐熱水		3.0%	5	5	5	5	5	5	5	Saliva (DIN) 耐唾液			3.0%	4-5
Sea Water 耐海水		3.0%	5	5	4	5	5	5	5	Hot-Pressing 耐熱壓	Dry 乾	Immediately	3.0%	4-5
Post Mercerizing 後絲光		3.0%	2-3R		4						4 hours after	5		
Acid Hydrolysis 酸水解		3.0%	4-5	5	4-5	4-5	5	5	4-5		Wet 濕	Immediately	3.0%	4-5
Perspiration 耐汗	Acid 酸	3.0%	5	5	4-5	4-5	5	5	5			4 hours after	4-5	
	Alkali 鹼		5	5	4-5	4-5	5	5	5	Nitrogen Oxides 氣體變褪色			3.0%	4-5

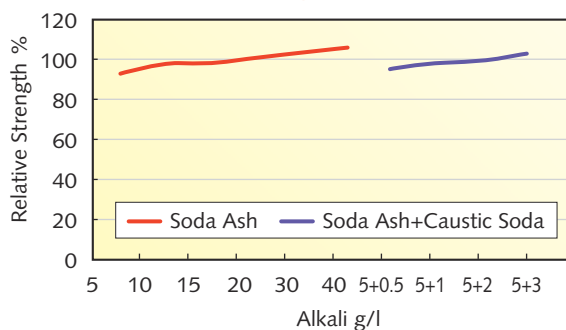
Light Fastness 日光堅牢度

Evaluation	Dyeing Depth	Color Change	ISO Blue	Evaluation		Dying Depth	Color Change	Grade ISO
Fastness	染色濃度	20 AFU	Scale	Fastness		染色濃度	20 AFU	Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.1%	2-3	3-4	Perspiration Light 耐汗日光	Acid	3.0%	2	4
	0.3%	3	3-4		Alkali		2-3	3-4
	1.0%	3-4	4					
	3.0%	4	4-5					
	6.0%	4-5	5					

浴比依存性 Dependency on Liquor Ratio



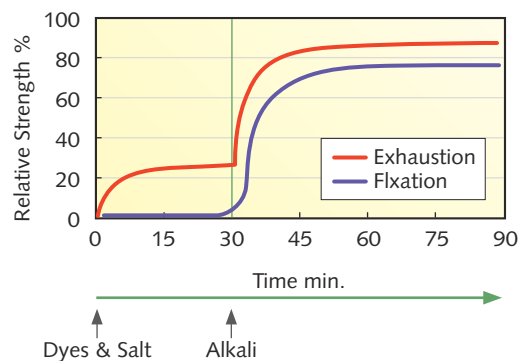
鹼依存性 Dependency on Alkali



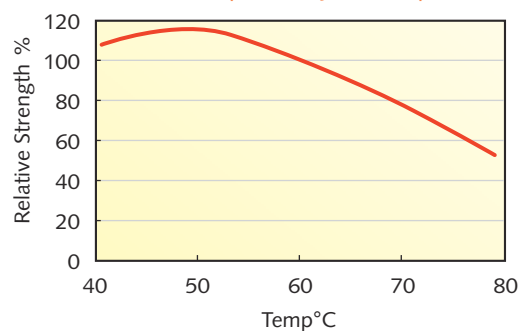
Dyeing Properties 特性

Characteristics		Evaluation
Reactivity 反應性		II
Substantivity 親和性		C
Migration 移染性		4-5
Washing Off 洗淨性		3
Solubility g/l 溶解度	50°C Water	100
	50°C Water + 50 g/l Salt	100
High Temp. Dyeing 高溫染色性	Dyed Fabric 色布	3-4
	Dye Liquor 染浴	3-4R
Color Constancy 色恆性	Tungsten Lamp	G(Y)
	TL 84 Lamp	G(Y)
Metallic Ions 金屬離子	Fe ²⁺ 鐵	4-5
	Cu ²⁺ 銅	2-3
Resin Finishing 樹脂加工	Melamine 三聚氰胺	4-5
	Glyoxal 乙二醛	4
Shade Change after Fixing Treatment 固色後變色性	Polyamine 聚胺類	4
	Quaternary Ammonium Salt 四級銨鹽	4
Stripping 脫色性	(1)	X
	(2)	△
	(3)	△
	(4)	○
Discharge Ability 拔染性	Neutral 中性	3-4
	Alkaline 鹼性	4

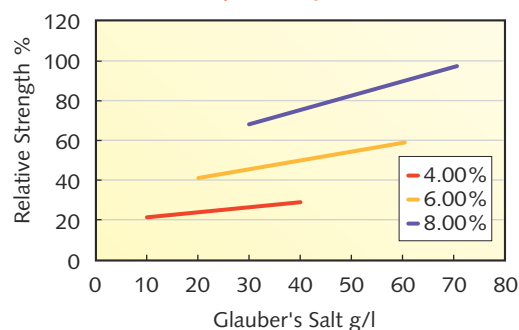
吸盡固著曲線 Exhaustion & Fixation



溫度依存性 Dependency on Temperature



芒硝依存性 Dependency on Glauber's Salt



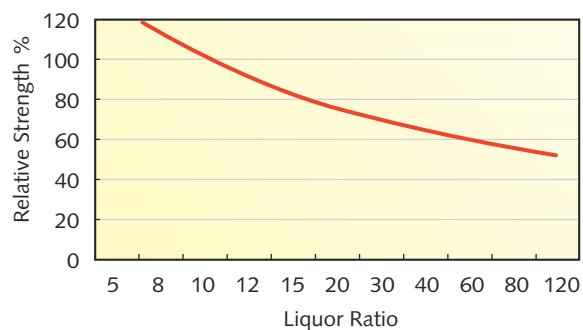
Fastness before Fixing Treatment 固色前牢度表

Evaluation		Depth of Shade 染色濃度	Color Change 變退色	Staining 污染						Evaluation			Depth of Shade 染色濃度	Grade 級數
Fastness				Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛	Fastness				
Washing 耐水洗	ISO C10 B	3.0%	4-5	5	5	5	5	5	5	Chlorinated Water 耐氯水 (20ppm)			3.0%	4
	ISO C10 C		4-5	5	4-5	4-5	5	5	5	Peroxide Bleaching 雙氧水漂白				4-5
	ISO C10 D		4-5	5	4	5	5	5	4	Rubbing 耐摩擦	Dry 乾	3.0%	5	
	ISO C06-C1S		4-5	5	4-5	5	5	5	5		Wet 濕		2-3	
	ISO C06-C2S		4-5	5	5	5	5	5	5	M&S C9A 擬濕日光 (Detergent Light)			0.3%	4-5
	M&S C4A		4-5	5	5	5	5	5	5				3.0%	4-5
Water 耐水		3.0%	4-5	5	5	5	5	5	5	Multi-Washing 耐多次水洗			3.0%	4
Hot Water 耐熱水		3.0%	4-5	5	5	5	5	5	5	Saliva (DIN) 耐唾液			3.0%	5
Sea Water 耐海水		3.0%	4-5	5	5	5	5	5	5	Hot-Pressing 耐熱壓	Dry 乾	Immediately	3.0%	2-3
Post Mercerizing 後絲光		3.0%	3		4							4 hours after		4-5
Acid Hydrolysis 酸水解		3.0%	4-5	5	4-5	4	5	5	4-5		Wet 濕	Immediately	3.0%	2-3
Perspiration 耐汗	Acid 酸	3.0%	4-5	5	5	5	5	5	5			4 hours after		4-5
	Alkali 鹼		4-5	5	5	5	5	5	5	Nitrogen Oxides 氣體變褪色			3.0%	4-5

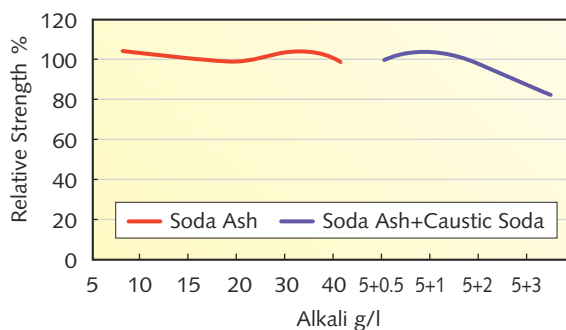
Light Fastness 日光堅牢度

Evaluation	Dyeing Depth	Color Change	ISO Blue	Evaluation		Dyeing Depth	Color Change	Grade ISO
Fastness	染色濃度	20 AFU	Scale	Fastness		染色濃度	20 AFU	Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.1%	1-2	2-3	Perspiration Light 耐汗日光	Acid	3.0%	2-3	4
	0.3%	2-3	3-4		Alkali		3	3-4
	1.0%	3	4					
	3.0%	4	5					
	6.0%	4-5	6					

浴比依存性 Dependency on Liquor Ratio



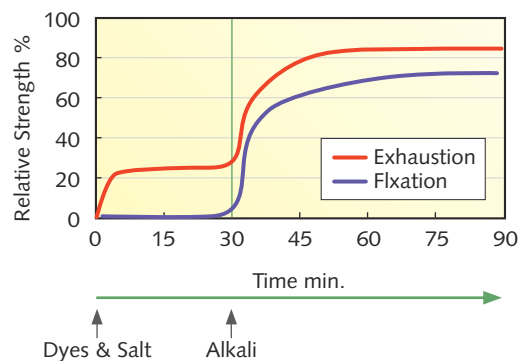
鹼依存性 Dependency on Alkali



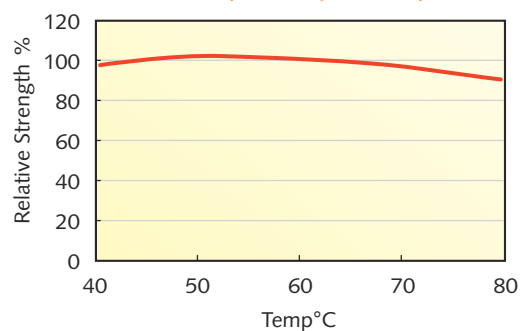
Dyeing Properties 特性

Characteristics		Evaluation
Reactivity 反應性		II
Substantivity 親和性		C
Migration 移染性		5
Washing Off 洗淨性		4-5
Solubility g/l 溶解度	50°C Water	150
	50°C Water + 50 g/l Salt	150
High Temp. Dyeing 高溫染色性	Dyed Fabric 色布	2-3
	Dye Liquor 染浴	2-3
Color Constancy 色恆性	Tungsten Lamp	R
	TL 84 Lamp	R
Metallic Ions 金屬離子	Fe ²⁺ 鐵	4
	Cu ²⁺ 銅	4
Resin Finishing 樹脂加工	Melamine 三聚氰胺	4-5
	Glyoxal 乙二醛	4-5
Shade Change after Fixing Treatment 固色後變色性	Polyamine 聚胺類	5
	Quaternary Ammonium Salt 四級銨鹽	5
Stripping 脫色性	(1)	△
	(2)	○
	(3)	○
	(4)	○
Discharge Ability 拔染性	Neutral 中性	3-4
	Alkaline 鹼性	4-5

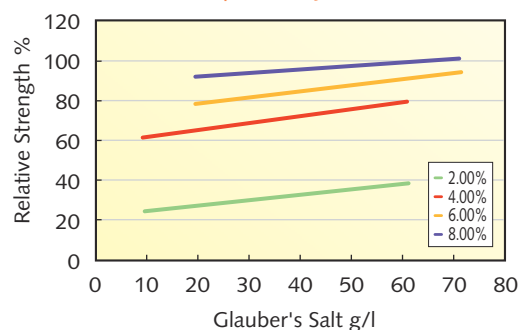
吸盡固著曲線 Exhaustion & Fixation



溫度依存性 Dependency on Temperature



芒硝依存性 Dependency on Glauber's Salt



Everzol Black ERC

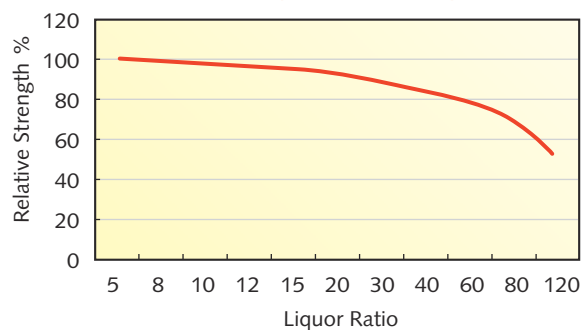
Fastness before Fixing Treatment 固色前牢度表

Evaluation		Depth of Shade 染色濃度	Color Change 變退色	Staining 污染						Evaluation			Depth of Shade 染色濃度	Grade 級數
Fastness				Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛	Fastness				
Washing 耐水洗	ISO C10 B	6.0%	4-5	5	5	4-5	5	5	4-5	Chlorinated Water 耐氯水 (20ppm)			6.0%	4-5
	ISO C10 C		4-5	5	4-5	5	5	5	4	Peroxide Bleaching 雙氧水漂白				5
	ISO C10 D		4-5	5	3	3	5	4-5	2	Rubbing 耐摩擦	Dry 乾	6.0%	4	
	ISO C06-C1S		4-5	5	5	4-5	5	5	4-5		Wet 濕		2-3	
	ISO C06-C2S		4-5	5	5	4-5	5	5	5	M&S C9A 擬濕日光 (Detergent Light)			1.0%	4
	M&S C4A		4-5	5	5	5	5	5	5				6.0%	>4-5
Water 耐水		6.0%	4-5	5	5	5	5	5	5	Multi-Washing 耐多次水洗			6.0%	4-5
Hot Water 耐熱水		6.0%	4-5	5	5	5	5	5	5	Saliva (DIN) 耐唾液			6.0%	5
Sea Water 耐海水		6.0%	4-5	5	5	5	5	5	5	Hot-Pressing 耐熱壓	Dry 乾	Immediately	6.0%	4-5
Post Mercerizing 後絲光		6.0%	2-3R		4-5							4 hours after		4-5
Acid Hydrolysis 酸水解		6.0%	4-5	5	5	4-5	5	5	5		Wet 濕	Immediately	6.0%	4-5
Perspiration 耐汗	Acid 酸	6.0%	4-5	5	5	5	5	5	5			4 hours after		4-5
	Alkali 鹼		4-5	5	5	5	5	5	5	Nitrogen Oxides 氣體變褪色			6.0%	5

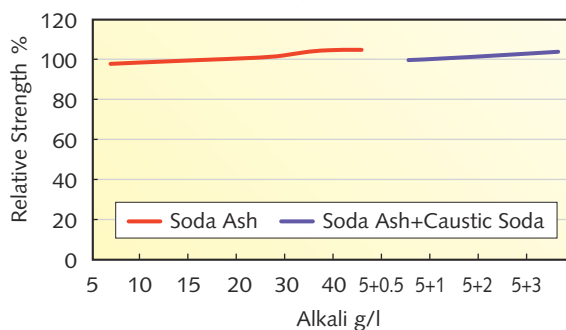
Light Fastness 日光堅牢度

Evaluation		Dyeing Depth 染色濃度	Color Change 20 AFU	ISO Blue Scale	Evaluation		Dying Depth 染色濃度	Color Change 20 AFU	Grade ISO Blue Scale
Fastness					Fastness				
Artificial Light (Xenon-Arc Lamp) 耐日光	0.3%	*	*	Perspiration Light 耐汗日光	Acid	6.0%	2	5	
	1.0%	2-3	4		Alkali		3	3-4	
	3.0%	3	4-5						
	6.0%	3-4	5-6						
	8.0%	4	5-6						

浴比依存性 Dependency on Liquor Ratio



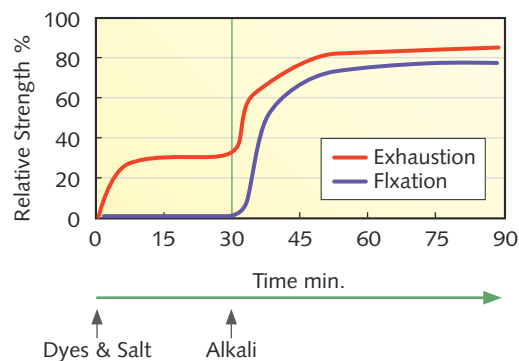
鹼依存性 Dependency on Alkali



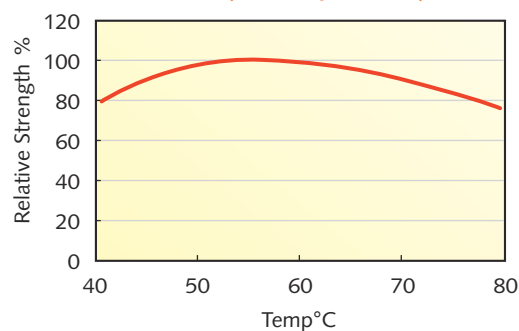
Dyeing Properties 特性

Characteristics		Evaluation
Reactivity 反應性		II
Substantivity 親和性		C
Migration 移染性		3-4
Washing Off 洗淨性		4-5
Solubility g/l 溶解度	50°C Water	100
	50°C Water + 50 g/l Salt	100
High Temp. Dyeing 高溫染色性	Dyed Fabric 色布	4-5
	Dye Liquor 染浴	4-5
Color Constancy 色恆性	Tungsten Lamp	R(B)
	TL 84 Lamp	G(B)
Metallic Ions 金屬離子	Fe ²⁺ 鐵	4-5
	Cu ²⁺ 銅	4-5
Resin Finishing 樹脂加工	Melamine 三聚氰胺	4-5
	Glyoxal 乙二醛	4-5
Shade Change after Fixing Treatment 固色後變色性	Polyamine 聚胺類	4-5
	Quaternary Ammonium Salt 四級銨鹽	4-5
Stripping 脫色性	(1)	△
	(2)	○
	(3)	○
	(4)	○
Discharge Ability 拔染性	Neutral 中性	3
	Alkaline 鹼性	4

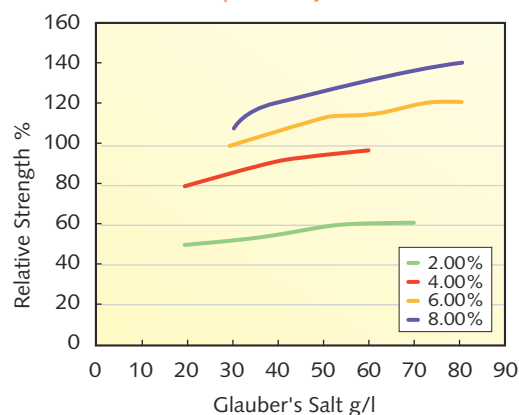
吸盡固著曲線 Exhaustion & Fixation



溫度依存性 Dependency on Temperature



芒硝依存性 Dependency on Glauber's Salt



Everzol Jet Black ERC

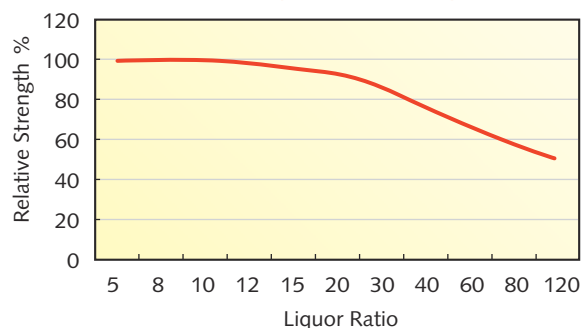
Fastness before Fixing Treatment 固色前牢度表

Evaluation		Depth of Shade 染色濃度	Color Change 變退色	Staining 污染						Evaluation			Depth of Shade 染色濃度	Grade 級數
Fastness				Acetate 醋酸纖維	Cotton 棉	Nylon 尼龍	Polyester 聚酯	Acrylic 壓克力	Wool 羊毛	Fastness				
Washing 耐水洗	ISO C10 B	6.0%	5	5	5	5	5	5	5	Chlorinated Water 耐氯水 (20ppm)			6.0%	4-5
	ISO C10 C		4-5	5	5	5	5	5	5	Peroxide Bleaching 雙氧水漂白			4-5	
	ISO C10 D		4-5	5	2-3	5	5	5	3	Rubbing 耐摩擦	Dry 乾	6.0%	4-5	
	ISO C06-C1S		5	5	5	5	5	5	5		Wet 濕	2-3		
	ISO C06-C2S		5	5	5	5	5	5	5	M&S C9A 擬濕日光 (Detergent Light)			1.0%	4
	M&S C4A		4-5	5	5	5	5	5	5				6.0%	4-5
Water 耐水		6.0%	4-5	5	4-5	4-5	5	5	5	Multi-Washing 耐多次水洗			6.0%	4-5
Hot Water 耐熱水		6.0%	4-5	5	5	5	5	5	5	Saliva (DIN) 耐唾液			6.0%	5
Sea Water 耐海水		6.0%	4-5	5	5	5	5	5	5	Hot- Pressing 耐熱壓	Dry 乾	Immediately	6.0%	4-5
Post Mercerizing 後絲光		6.0%	3-4		3							4 hours after	4-5	
Acid Hydrolysis 酸水解		6.0%	4-5	5	4-5	5	5	5	5		Wet 濕	Immediately	6.0%	4-5
Perspiration 耐汗	Acid 酸	6.0%	4-5	5	5	5	5	5	5			4 hours after	4-5	
	Alkali 鹼		4-5	5	4-5	4	5	5	5	Nitrogen Oxides 氣體變褪色			6.0%	5

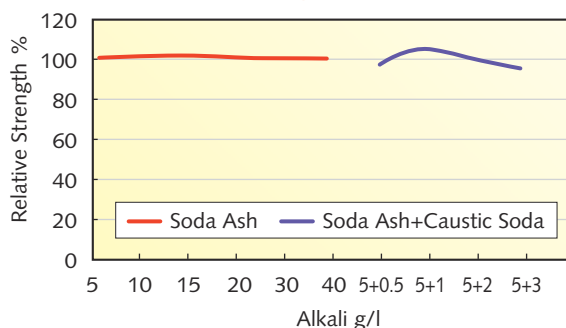
Light Fastness 日光堅牢度

Evaluation	Dyeing Depth	Color Change	ISO Blue	Evaluation		Dying Depth	Color Change	Grade ISO
Fastness	染色濃度	20 AFU	Scale	Fastness		染色濃度	20 AFU	Blue Scale
Artificial Light (Xenon-Arc Lamp) 耐日光	0.3%	2	4	Perspiration Light 耐汗日光	Acid	6.0%	3	5
	1.0%	3	4-5		Alkali		3-4	4
	3.0%	3-4	5					
	6.0%	4	5-6					
	8.0%	4-5	6					

浴比依存性 Dependency on Liquor Ratio



鹼依存性 Dependency on Alkali



Note

[illegible]

Note

[illegible]

本資料所提供的資訊與建議是以我們最大的用心完成，但不能含括所有可能的狀況。使用相關產品或搭配其他產品時，請先自行測試。本資料並不具保證及法律效用，僅供參考使用。使用者必須自行承擔所有的風險與法律責任。

臺灣永光化學工業股份有限公司－技術行銷處

Warnings: The information and recommendations presented here were completed with the utmost care, but can't be extended to cover every possible case. They are intended to serve as non-binding guidelines and must be adapted to the prevailing condition. The product whether used singly or in combination with other substances, should be carried out in lab test prior to bulk production. The user needs to assume all risk and liability no matter whatsoever results are.

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